

CONDUCTION

Fourier Law of Conduction

Rate of heat conduction is proportional to the area measured normal to the direction of heat flow and to the temperature gradient in that direction.

$$Q \propto A \frac{dT}{dx}$$

$$Q = -kA \frac{dT}{dx}$$

Where

A – Area in m²

$\frac{dT}{dx}$ – Temperature gradient in K/m

k – Thermal conductivity in W/mK

Thermal conductivity is defined as the ability of a substance to conduct heat.

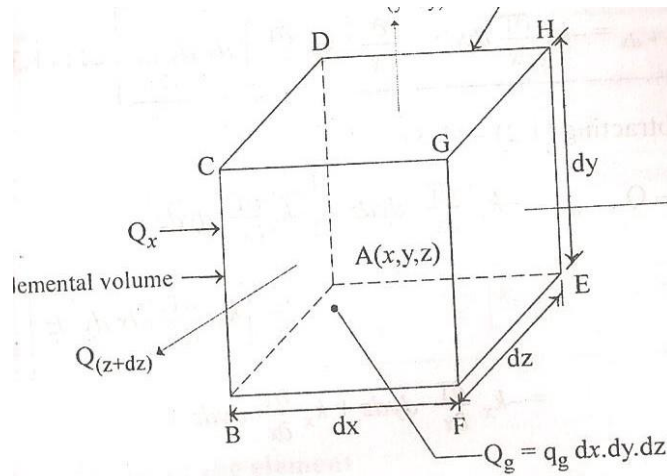
[The negative sign indicates that the heat flows in a direction along which there is a decrease in temperature]

GENERAL HEAT CONDUCTION EQUATION IN CARTESIAN COORDINATES

Consider a small rectangular element of sides d_x , d_y and d_z as shown in Fig. 1.1.

The energy balance of this rectangular element is obtained from first law of thermodynamics.

Net heat conducted into element from all the coordinate directions	+	Heat generated within the element	=	Heat stored stored in the element	..(1.1)
					



Net heat conducted into element from all the coordinate directions

Let q_x be the heat flux in a direction of face ABCD and $q_x + d_x$ be the heat flux in a direction of face EFGH.

The rate of heat flow into the element in x direction through the face ABCD is

$$Q_x = q_x dydz = -k_x \frac{\partial T}{\partial x} dydz \quad \dots (1.2)$$

Where k – Thermal conductivity, W/mK

$\frac{\partial T}{\partial x}$ – Temperature gradient

The rate of heat flow out of the element in x direction through the face EFGH is

$$Q_x + d_x = -Q_x + \frac{\partial}{\partial x}(Q_x)dx$$

$$= -k_x \frac{\partial T}{\partial x} dydz + \frac{\partial}{\partial x} \left[-k_x \frac{\partial T}{\partial x} dydz \right] dx$$

$$Q_x + d_x = -k_x \frac{\partial T}{\partial x} dydz - \frac{\partial}{\partial x} \left[k_x \frac{\partial T}{\partial x} \right] dx dydz$$

Subtracting (1.2) – (1.3)

$$\begin{aligned}
 Q - d_{(x+dx)} &= -k \frac{\partial T}{\partial x} dydz - \left[k \frac{\partial T}{\partial x} dydz - \frac{\partial}{\partial x} \left[k \frac{\partial T}{\partial x} \right] dx dy dz \right] \\
 &= -k \frac{\partial T}{\partial x} dydz + k \frac{\partial T}{\partial x} dydz + \frac{\partial}{\partial x} \left[k \frac{\partial T}{\partial x} \right] dx dy dz \\
 \Rightarrow Q - Q_{(x+dx)} &= \frac{\partial}{\partial x} \left[k \frac{\partial T}{\partial x} \right] dx dy dz \quad \dots (1.4)
 \end{aligned}$$

Similarly

$$Q + d_{(y+dy)} = \frac{\partial}{\partial y} \left[k \frac{\partial T}{\partial y} \right] dx dy dz \quad \dots (1.5)$$

$$QZ - q_{(y+dy)} = \frac{\partial}{\partial y} \left[k \frac{\partial T}{\partial y} \right] dx dy dz \quad \dots (1.6)$$

ADDING (1.4) + (1.5) + (1.6)

$$\begin{aligned}
 \text{Net heat conducted} &= \frac{\partial}{\partial x} \left[k \frac{\partial T}{\partial x} \right] dx dy dz + \frac{\partial}{\partial y} \left[k \frac{\partial T}{\partial y} \right] dx dy dz + \frac{\partial}{\partial z} \left[k \frac{\partial T}{\partial z} \right] dx dy dz \\
 &= \left[\frac{\partial}{\partial x} \left[k \frac{\partial T}{\partial x} \right] + \frac{\partial}{\partial y} \left[k \frac{\partial T}{\partial y} \right] + \frac{\partial}{\partial z} \left[k \frac{\partial T}{\partial z} \right] \right] dx dy dz
 \end{aligned}$$

Net heat conducted into element from all the coordinate directions

$$= \left[\frac{\partial}{\partial x} \left[k \frac{\partial T}{\partial x} \right] + \frac{\partial}{\partial y} \left[k \frac{\partial T}{\partial y} \right] + \frac{\partial}{\partial z} \left[k \frac{\partial T}{\partial z} \right] \right] dx dy dz$$

Heat Stored in the element

We know that,

$$\begin{array}{ccccccc}
 \text{Heat stored in the} & & \text{Mass of the} & & \text{Specific heat of} & & \text{Rise in} \\
 \text{element} & = & \text{element} & \times & \left. \begin{array}{c} \text{Heat of the} \\ \text{element} \end{array} \right\} & \times & \left. \begin{array}{c} \text{temperature} \\ \text{of element} \end{array} \right\}
 \end{array}$$

$$= m \times C_p \times \frac{\partial T}{\partial t}$$

$$= \rho \times d_x d_y d_z \times C_p \times \frac{\partial T}{\partial t}$$

[$\because$ Mass = Density $\times$ Volume]

Heat stored in the element = $\rho C_p \frac{\partial T}{\partial t} d_x d_y d_z$	 (1.9)
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Heat Stored within the element

Heat generated within the element is given by

$$Q = q d_x d_y d_z \quad \text{..... (1.9)}$$

Substituting equation (1.7), (1.8) and (1.9) in equation (1.1)

$$\begin{aligned} (1.1) \quad & \Rightarrow \left[\frac{\partial}{\partial x} \left[k \frac{\partial T}{\partial x} \right] + \frac{\partial}{\partial y} \left[k \frac{\partial T}{\partial y} \right] + \frac{\partial}{\partial z} \left[k \frac{\partial T}{\partial z} \right] \right] d_x d_y d_z + q d_x d_y d_z = \rho C_p \frac{\partial T}{\partial t} d_x d_y d_z \\ & \Rightarrow \frac{\partial}{\partial x} \left[k_x \frac{\partial T}{\partial x} \right] + \frac{\partial}{\partial y} \left[k_y \frac{\partial T}{\partial y} \right] + \frac{\partial}{\partial z} \left[k_z \frac{\partial T}{\partial z} \right] + q = \rho C_p \frac{\partial T}{\partial t} \end{aligned}$$

Considering the material is isotropic. So,

$$k_x = k_y = k_z = k = \text{constant.}$$

$$\Rightarrow \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right] k + q = \rho C_p \frac{\partial T}{\partial t}$$

Divided by k,

$$\begin{aligned} \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{q}{k} &= \frac{\rho C_p}{k} \frac{\partial T}{\partial t} \\ \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{q}{k} &= \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \text{..... (1.10)} \end{aligned}$$

It is a general three dimensional heat conduction equation in Cartesian coordinates

Where, $\alpha = \text{Thermal diffusivity} = \frac{k}{\rho C_p} - m^2/s$

Thermal diffusivity is nothing but how fast heat is diffused through a material during changes of temperature with time.

Case (i): No heat sources

In the absence of internal heat generation, equation (1.10) reduces to

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \dots (1.11)$$

This equation is known as diffusion equation (or) Fourier's equation.

Case (ii): Steady state conditions

In steady state condition, the temperature does not change with time. So, $\frac{\partial T}{\partial t} = 0$. The heat conduction equation (1.10) reduces to

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = \frac{q}{k} = 0 \quad \dots (1.12)$$

(or)

$$\nabla^2 T + \frac{q}{k} = 0$$

This equation is known as known as Poisson's equation.

In the absence of internal heat generation, equation (1.12) becomes:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0 \quad \dots (1.13)$$

(or)

$$\nabla^2 T = 0$$

This equation is known as Laplace equation.

Case (iii): One dimensional steady state heat conduction

If the temperature varies only in the x direction, the equation (1.10) reduces to

$$\frac{\partial^2 T}{\partial x^2} + \frac{q}{k} = 0 \quad \dots\dots (1.14)$$

In the absence of internal heat generation, equation (1.14) becomes:

$$\frac{\partial^2 T}{\partial x^2} = 0 \quad \dots\dots (1.15)$$

Case (iv): Two dimensional steady state heat conduction

If the temperature varies only in the x and y direction, the equation (1.10) becomes:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{q}{k} = 0 \quad \dots\dots (1.16)$$

In the absence of internal heat generation, equation (1.16) reduces to

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0 \quad \dots\dots (1.17)$$

Case (v): Unsteady state, one dimensional, without internal heat generation

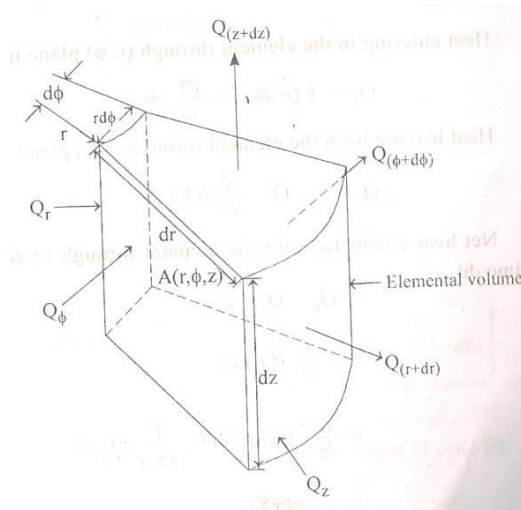
In unsteady state, the temperature changes with time, i.e., $\frac{\partial T}{\partial t} \neq 0$. So, the general conduction equation (1.10) reduces to

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \dots\dots (1.18)$$

GENERAL HEAT CONDUCTION EQUATION IN CYLINDRICAL CO-ORDINATES

The general heat conduction equation in Cartesian coordinates derived in the previous section is used for solids with rectangular boundaries like squares, cubes, slabs etc. But, the Cartesian coordinate system is not applicable for the solids like cylinders, cones, spheres etc. for cylindrical solids, a cylindrical coordinate system is used.

Consider a small cylindrical element of sides dr , $d\phi$ and dz as shown in fig. 1.2.



The volume of the element $dv = r \, d\phi \, dr \, dz$.

Let us assume that thermal conductivity k , Specific heat C_p and density ρ are constant.

The energy balance of this cylindrical element is obtained from first law of thermodynamics.

Net heat conducted into element from all the coordinate directions	+	Heat generated within the element	=	Heat stored stored in the element	..(1.19)
$\left. \begin{array}{l} \text{Net heat conducted into element from all the co-ordinate directions} \\ \text{Heat entering in the element through } (r, \phi) \text{ plane in time } d\theta \end{array} \right\} = \left\{ \begin{array}{l} \text{Heat stored} \\ \text{stored in the} \\ \text{element} \end{array} \right. \quad \text{..(1.19)}$					

$$Q_z = -k (r d\phi dr) \frac{\partial T}{\partial z} d\theta$$

Heat leaving from the element through (r, ϕ) plane in time $d\theta$.

$$Q_{z+dz} = Q_z + \frac{\partial}{\partial z}(Q_z) dz$$

Net heat conducted into the element through (r, ϕ) plane in time $d\theta$.

$$\begin{aligned}
&= Q_z - Q_{z+dz} \\
&= -\frac{\partial}{\partial z}(Q_z)dz \\
&= \frac{\partial}{\partial z} \left[k(r d\phi \cdot dr) \left[\frac{\partial T}{\partial z} \right] d\theta \right] dz \\
&= k \left[\frac{\partial^2 T}{\partial z^2} \right] (dr \cdot r d\phi \cdot dz) d\theta
\end{aligned}$$

Net heat conducted through (r, ϕ) plane $= \left[\frac{\partial^2 T}{\partial z^2} \right] (dr \cdot r d\phi \cdot dz) d\theta$

..... (1.20)

Heat entering in the element through (ϕ, z) plane in time $d\theta$.

$$\begin{aligned}
&= Q_r - Q_{r+dr} \\
&= -\frac{\partial}{\partial r}(Q_r)dr \\
&= -\frac{\partial}{\partial r} \left[k(r d\phi \cdot dz) \left[\frac{\partial T}{\partial r} \right] d\phi \right] dr \\
&= k(drd\phi \cdot dz) \cdot \frac{\partial}{\partial r} \left[\frac{1}{r} \frac{\partial T}{\partial r} \right] d\theta \\
&= k(dr \cdot r d\phi \cdot dz) \left[\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} \right] d\theta
\end{aligned}$$

Net heat conducted through (ϕ, z) plane $= k(dr \cdot r d\phi \cdot dz) \left[\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} \right] d\theta$

Heat entering in the element through (z, r) plane in time $d\theta$.

$$Q_\phi = -k(dr \cdot dz) \frac{\partial T}{r \partial \phi} d\theta$$

Heat leaving from the element through (z, r) plane in time $d\theta$.

$$Q_{\phi+d\phi} = Q_\phi + \frac{\partial}{\partial \phi}(Q_\phi) r d\phi$$

Net heat conducted into the element through (z, r) plane in time $d\theta$.

$$\begin{aligned}
 Q_{\phi} - Q_{\phi+d\phi} &= - \frac{\partial}{\partial \phi} (Q_{\phi}) r d\phi \\
 &= - \frac{\partial}{\partial \phi} \left[k (dr dz) \frac{\partial T}{\partial \phi} \right] r d\phi \\
 &= k \frac{\partial}{\partial \phi} \left[r \frac{\partial T}{\partial \phi} \right] dr dz d\theta \\
 &= k \left[\frac{1}{r^2} \frac{\partial^2 T}{\partial \phi^2} \right] (r dr dz d\theta)
 \end{aligned}$$

Net heat conducted into element from all the co-ordinate directions

$$\begin{aligned}
 &= k \frac{\partial^2 T}{\partial z^2} (dr rd\phi dz) d\theta \\
 &+ \\
 &k (dr rd\phi dz) \left[\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} \right] d\theta \\
 &k \left[\frac{1}{r^2} \frac{\partial^2 T}{\partial \phi^2} \right] (r dr dz d\theta)
 \end{aligned}$$

[Adding equation 1.20, 1.21 and 1.22]

$$\begin{aligned}
 &= k (dr rd\phi dz) d\theta \left[\frac{\partial^2 T}{\partial z^2} + \frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial^2 T}{\partial \phi^2} \right] \\
 &= k (dr rd\phi dz) d\theta \left[\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2} \right]
 \end{aligned}$$

Net heat conducted into element from all the co-ordinate directions

$$= k (dr rd\phi dz) d\theta \left[\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2} \right]$$

..... (1.23)

Heat generated within the element

Total heat generated within the element is given by

$$Q = q (dr \, rd\phi \, dz) \, d\theta \quad \dots\dots (1.24)$$

Heat stored in the element

The increase in internal energy of the element is equal to the net heat stored in the element.

Increase in internal energy

= Net heat stored in the element

$$= \rho (dr \, rd\phi \, dz) C_p \frac{\partial T}{\partial \theta} \times d\theta \quad \dots\dots (1.25)$$

Substituting equation (1.23), (1.24) and (1.25) in (1.19)

$$\begin{aligned} (1.19) \Rightarrow & k (dr \, rd\phi \, dz) d\theta \left[\frac{\partial^2 T}{\partial r^2} + r \frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2} \right] + q (dr \, rd\phi \, dz) C_p \frac{\partial T}{\partial \theta} \times d\theta \\ & = \rho (dr \, rd\phi \, dz) C_p \frac{\partial T}{\partial \theta} \times d\theta \end{aligned}$$

Divided by $(dr \, rd\phi \, dz) \, d\theta$

$$\begin{aligned} \Rightarrow & k \left[\frac{\partial^2 T}{\partial r^2} + r \frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2} \right] + q = \rho C_p \frac{\partial T}{\partial \theta} \\ \Rightarrow & \boxed{\left[\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2} \right] + \frac{q}{k} = \frac{\rho C_p}{k} \frac{\partial T}{\partial \theta}} \quad \dots\dots (1.26) \end{aligned}$$

It is a general three dimensional heat conduction equation in cylindrical co-ordinates.

$$\Rightarrow \frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2} + \frac{q}{k} = \frac{\rho C_p}{k} \frac{\partial T}{\partial \theta}$$

If the flow is steady, one dimensional and no heat generation, equation (1.26) becomes:

$$\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} = 0 \quad \dots\dots (1.27)$$

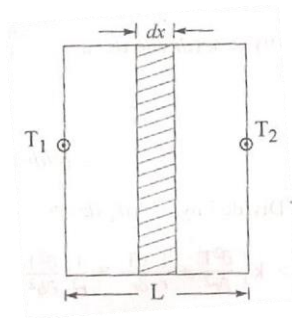
(or)

$$\implies \frac{1}{r} + \frac{d}{dr} \left[r \cdot \frac{\partial T}{\partial r} \right] = 0 \quad \dots\dots (1.28)$$

CONDUCTION OF HEAT THROUGH A SLAB OR PLANE WALL

Consider a slab of uniform thermal conductivity k , thickness L , with inner temperature T_1 , and outer temperature T_2 .

Let us consider a small elemental area of thickness „ dx “.



From Fourier law of conduction, we know that,

$$Q = -kA \frac{dT}{dx}$$

$$\implies Q \cdot dx = -kA dT$$

Integrating the above equation between the limits of 0 to L and T_1 to T_2 .

$$\implies \int_0^L Q dx = - \int_{T_1}^{T_2} kA dT$$

$$\implies Q \int_0^L dx = -kA \int_{T_1}^{T_2} dT$$

$$\implies Q [x]_0^L = -kA [T]_{T_1}^{T_2}$$

$$\implies Q [L - 0] = -kA [T_2 - T_1]$$

$$\implies Q \times L = kA [T_1 - T_2]$$

$$\implies Q = \frac{kA}{L} [T_1 - T_2] \quad \dots\dots (1.29)$$

$$\begin{aligned} \Rightarrow Q &= \frac{T_1 - T_2}{\frac{L}{kA}} \\ \Rightarrow Q &= \frac{\Delta T_{\text{overall}}}{R} \end{aligned} \quad \dots\dots (1.30)$$

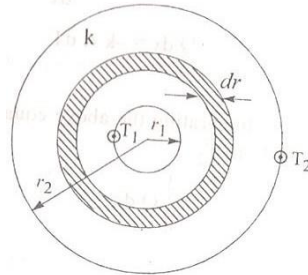
Where

$$\Delta T = T_1 - T_2$$

$$R = \frac{L}{kA} - \text{Thermal resistance of slab.}$$

CONDUCTION OF HEAT THROUGH HOLLOW CYLINDER

Consider a hollow cylinder inner radius r_1 , outer radius r_2 , inner temperature T_1 , outer temperature T_2 and thermal conductivity k .



Let us consider a small elemental area of thickness “dr”

From Fourier law of conduction, we know that,

$$Q = -kA \frac{dT}{dr}$$

Area of a cylinder is $2\pi rL$

$$A = 2\pi rL$$

So,

$$Q = -k 2\pi r L \frac{dT}{dr}$$

$$\Rightarrow Q \times \frac{dr}{r} = -k 2\pi L dT$$

Integrating the above equation from r_1 to r_2 and T_1 to T_2 .

$$\Rightarrow Q \int_{r_1}^{r_2} \frac{dr}{r} = -k 2\pi L \int_{T_1}^{T_2} dT$$

$$\Rightarrow Q [In r]_{r_1}^{r_2} = -k 2\pi L [T]_{T_1}^{T_2}$$

$$\Rightarrow Q [In r_2 - In r_1] = -k 2\pi L [T_2 - T_1]$$

$$\Rightarrow Q In \left(\frac{r_2}{r_1} \right) = 2\pi L k [T_1 - T_2]$$

$$\Rightarrow Q = \frac{2\pi L k [T_1 - T_2]}{In \left(\frac{r_2}{r_1} \right)} \quad \dots (1.31)$$

$$\Rightarrow Q = \frac{T_1 - T_2}{\frac{1}{2\pi L k In \left(\frac{r_2}{r_1} \right)}}$$

$$\Rightarrow Q = \frac{\Delta T_{overall}}{R} \quad \dots (1.32)$$

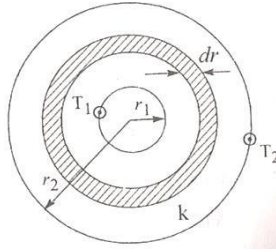
Where

$$\Delta T = T_1 - T_2$$

$$R = \frac{1}{2\pi L k In \left(\frac{r_2}{r_1} \right)} - \text{Thermal resistance of the hollow cylinder.}$$

CONDUCTION OF HEAT THROUGH HOLLOW SPHERE

Consider a hollow sphere of inner radius r_1 , outer radius r_2 , inner temperature T_1 , outer temperature T_2 and thermal conductivity k .



Let us consider a small elemental area of thickness „dr“. From Fourier law of heat conduction, we know that

$$Q = -kA \frac{dT}{dr}$$

Area of sphere is $4\pi r^2$

$$A = 4\pi r^2$$

$$Q = -k4\pi r^2 \frac{dT}{dr} \quad \dots\dots (1.33)$$

$$\Rightarrow Q \times \frac{dr}{r^2} = -k \times 4\pi dT$$

Integrating on both sides

$$\Rightarrow Q \int_{r_1}^{r_2} \frac{dr}{r^2} = - \int_{T_1}^{T_2} 4\pi k dT$$

$$\Rightarrow Q \int_{r_1}^{r_2} \frac{dr}{r^2} = -4\pi k \int_{T_1}^{T_2} dT$$

$$\Rightarrow Q \left[-\frac{1}{r} \right]_{r_1}^{r_2} = -4\pi k [T]_{T_1}^{T_2}$$

$$\Rightarrow Q \left[\frac{1}{r_1} - \frac{1}{r_2} \right] = -4\pi k [T_2 - T_1]$$

$$\Rightarrow Q \left[\frac{r_2 - r_1}{r_1 r_2} \right] = 4\pi k [T_1 - T_2]$$

$$\Rightarrow Q = \frac{4\pi k [T_1 - T_2]}{\frac{r_2 - r_1}{r_1 r_2}} \quad \dots\dots (1.34)$$

$$\Rightarrow Q = \frac{T_1 - T_2}{\frac{r_2 - r_1}{4\pi k(r_1 r_2)}}$$

$$\Rightarrow Q = \frac{\Delta T_{overall}}{R} \quad \dots\dots\dots (1.35)$$

Where

$$\Delta T = T_1 - T_2$$

$$R = \frac{r_2 - r_1}{4\pi k(r_1 r_2)} - \text{Thermal resistance of hollow sphere.}$$

Newton's Law of Cooling

Heat transfer by convection is given by Newton law of cooling

$$Q = hA[T_s - T_\alpha] \quad \dots\dots\dots (1.36)$$

Where

A - Area exposed to heat transfer in m²

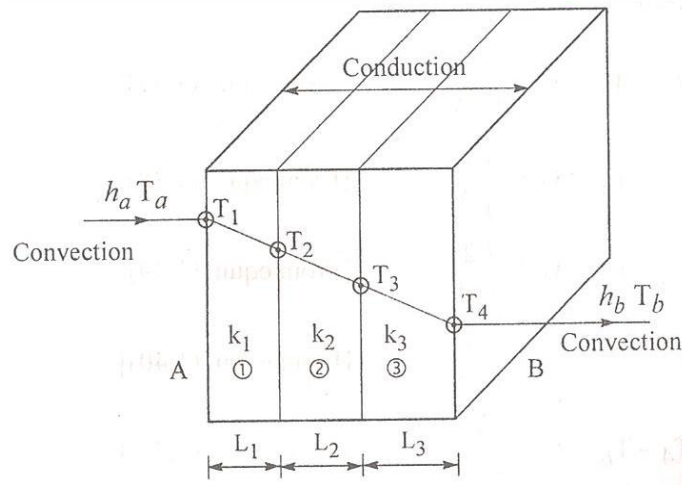
h – Heat transfer co-efficient in W/m²K

T_s – Temperature of the surface in K

T_α - Temperature of the fluid in K.

6. Heat Transfer Through a Composite Plane Wall with Inside and Outside Convection

Consider a composite wall of thickness L₁, L₂ and L₃ having thermal conductivity k₁, k₂ and k₃ respectively. It is assumed that the interior and exterior surface of the system are subjected to convection at mean temperatures T_a and T_b with heat transfer co-efficient h_a and h_b respectively. Within the composite wall, the slabs are subjected to conduction.



From Newton's law of cooling, we know that,

Heat transfer by convection at side A is

$$Q = haA[T_a - T_1] \text{ [From equn. (1.36)]} \quad \dots\dots (1.37)$$

Heat transfer by conduction at slab (1) is

$$Q = \frac{k_1 A [T_1 - T_2]}{L_1} \text{ [From equn. (1.29)]} \quad \dots\dots\dots (1.38)$$

Heat transfer by conduction at slab (2) is

$$Q = \frac{k_2 A [T_2 - T_1]}{L_2} \quad \dots\dots\dots (1.39)$$

Similarly at slab (3) is

$$Q = \frac{k_3 A [T_3 - T_4]}{L_3} \quad \dots\dots\dots (1.40)$$

Heat transfer by convection at side B is

$$Q = hbA[T_4 - T_b] \quad \dots\dots\dots (1.41)$$

We know that,

$$T_a - T_1 = Q \times \frac{1}{h_a A} \quad [\text{From equn. (1.37)}]$$

$$T_1 - T_2 = Q \times \frac{L_1}{k_1 A} \quad [\text{From equn. (1.38)}]$$

$$T_2 - T_3 = Q \times \frac{L_2}{k_2 A} \quad [\text{From equn. (1.39)}]$$

$$T_3 - T_4 = Q \times \frac{L_3}{k_3 A} \quad [\text{From equn. (1.40)}]$$

$$T_4 - T_b = Q \times \frac{1}{h_b A} \quad [\text{From equn. (1.410)}]$$

Adding both sides of the above equations

$$\Rightarrow T_a - T_b = Q \left[\frac{1}{h_a A} + \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \frac{L_3}{k_3 A} + \frac{1}{h_b A} \right]$$

$$\Rightarrow Q = \frac{T_a - T_b}{\frac{1}{h_a A} + \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \frac{L_3}{k_3 A} + \frac{1}{h_b A}}$$

$$\Rightarrow Q = \frac{\Delta T_{\text{overall}}}{R} \quad \dots\dots\dots (1.42)$$

Where

$$\Delta T = T_a - T_b$$

Thermal resistance, $R = R_a + R_1 + R_2 + R_3 + R_b$

$$= \frac{1}{h_a A} + \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \frac{L_3}{k_3 A} + \frac{1}{h_b A}$$

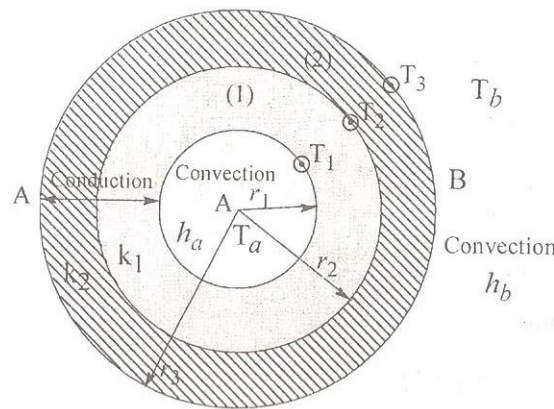
We know that,

$$\begin{aligned}
 R &= \frac{1}{UA} \\
 \Rightarrow Q &= \frac{T_a - T_b}{\frac{1}{UA}} \\
 \Rightarrow Q &= UA[T_a - T_b] \quad \dots\dots (1.43)
 \end{aligned}$$

Where

„U“ is the overall heat transfer co-efficient (W/m²K).

HEAT TRANSFER THROUGH COMPOSITE PIPES OR CYLINDERS WITH INSIDE AND OUTSIDE CONVECTION



A hot fluid at a temperature T_a , with heat transfer co-efficient h_a , flowing through a pipe is separated by two layers from atmosphere as shown in fig. 1.7. Let the thermal conductivities be k_1 and k_2 . On the outside surface heat is being transferred to a cold fluid at a temperature T_b with heat transfer co-efficient h_b .

Heat transfer by convection at side A is

$$Q = h_a A [T_a - T_1] \quad [\text{From equn. No. (1.36)}]$$

Here Area, $A = 2\pi r_1 L$

So,

$$Q = 2\pi r_1 L h_a [T_a - T_1] \quad \dots\dots (1.44)$$

Heat transfer by conduction at section 1 is

$$Q = \frac{2\pi L k_1 [T_1 - T_2]}{\ln \left(\frac{r_2}{r_1} \right)} \quad \dots\dots\dots (1.45)$$

[From equn. (1.310)]

Similarly

At section 2

$$Q = \frac{2\pi L k_2 [T_2 - T_3]}{\ln \left(\frac{r_3}{r_2} \right)} \quad \dots\dots\dots (1.46)$$

Heat transfer by convection at side B is

$$\begin{aligned} Q &= hbA[T_3 - T_b] \\ Q &= 2\pi r_3 L hb[T_3 - T_b] \end{aligned} \quad \dots\dots\dots (1.47)$$

We know that,

$$T_a - T_1 = \frac{Q}{2\pi L r_1 h_a} \quad \text{[From equn. (1.440)]}$$

$$T_1 - T_2 = \frac{Q}{2\pi L k_1} \ln \left(\frac{r_2}{r_1} \right) \quad \text{[From equn. (1.45)]}$$

$$T_2 - T_3 = \frac{Q}{2\pi L k_2} \ln \left(\frac{r_3}{r_2} \right) \quad \text{[From equn. (1.46)]}$$

$$T_3 - T_b = \frac{Q}{2\pi L r_3 hb} \quad \text{[From equn. (1.47)]}$$

Adding both sides of the above equations

$$T_a - T_b = \frac{Q}{2\pi L} \left[\frac{1}{h_a r_1} + \frac{\ln(r_2/r_1)}{k_1} + \frac{\ln(r_3/r_2)}{k_2} + \frac{1}{h_b r_3} \right]$$

$$\Rightarrow Q = \frac{2\pi L [T_a - T_b]}{\frac{1}{h_a r_1} + \frac{\ln(r_2/r_1)}{k_1} + \frac{\ln(r_3/r_2)}{k_2} + \frac{1}{h_b r_3}}$$

$$\Rightarrow Q = \frac{2\pi L [T_a - T_b]}{\left[\frac{1}{h_a r_1} + \frac{\ln(r_2/r_1)}{k_1} + \frac{\ln(r_3/r_2)}{k_2} + \frac{1}{h_b r_3} \right]}$$

$$Q = \frac{\Delta T_{overall}}{R}$$

Where

$$R = \frac{1}{2\pi L} \left[\frac{1}{h_a r_1} + \frac{\ln(r_2/r_1)}{k_1} + \frac{\ln(r_3/r_2)}{k_2} + \frac{1}{h_b r_3} \right]$$

$$\Delta T_{overall} = T_a - T_b$$

We know that,

$$R = \frac{1}{UA}$$

$$\Rightarrow Q = \frac{T_a - T_b}{\frac{1}{UA}}$$

$$Q = UA[T_a - T_b] \quad \dots\dots\dots (1.49)$$

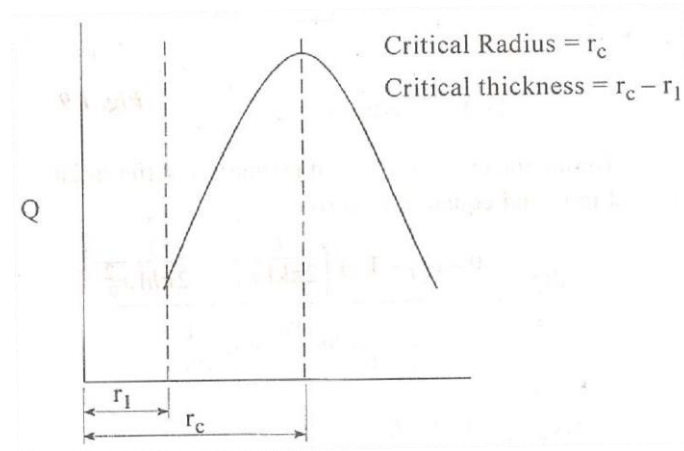
Where

U = Overall heat transfer co-efficient, W/m²K

A = Area = 2πr₃L

CRITICAL RADIUS OF INSULATION

Addition of insulating material on a surface does not reduce the amount of heat transfer rate always. In fact under certain circumstances it actually increases the heat loss up to certain thickness of insulation. The radius of insulation for which the heat transfer is maximum is called critical radius of insulation and the corresponding thickness is called critical thickness. If the thickness is further increased, the heat loss will be reduced.



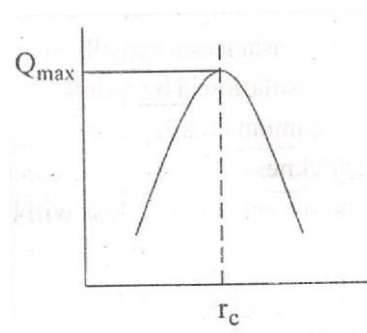
Critical Radius of Insulation For A Cylinder

Consider a cylinder having thermal conductivity k . Let r_1 and r_0 inner and outer radii of insulation.

$$\text{Heat transfer, } Q = \frac{T_i - T_\alpha}{\ln \frac{r_0}{r_1} \cdot \frac{2\pi k L}{r_1}} \quad [\text{From equn. No. (1.31)}]$$

Considering h be the outside heat transfer co-efficient.

$$\therefore Q = \frac{T_i - T_\alpha}{\ln \left(\frac{r_0}{r_1} \right) \cdot \frac{r_1}{2\pi k L} + \frac{1}{A_0 h}}$$



Here $A_0 = 2\pi r_0 L$

$$\Rightarrow Q = \frac{T_i - T_\alpha}{\frac{\ln\left(\frac{r_0}{r_1}\right)}{2\pi kL} + \frac{1}{2\pi_0 Lh}}$$

To find the critical radius of insulation, differentiate Q with respect to r_0 and equate it to zero.

$$\Rightarrow \frac{dQ}{dr_0} = \frac{0 - (T_i - T_\alpha) \left[\frac{1}{2\pi kL} - \frac{1}{2\pi hLr_0^2} \right]}{\frac{1}{2\pi kL} \ln\left(\frac{r_0}{r_1}\right) + \frac{1}{2\pi hLr_0}}$$

Since $(T_i - T_\alpha) \neq 0$

$$\Rightarrow \frac{1}{2\pi kLr_0} - \frac{1}{2\pi hLr_0^2} = 0$$

$$\Rightarrow r_0 = \frac{k}{h} = r_c$$

..... (1.50)

HEAT CONDUCTION WITH HEAT GENERATION

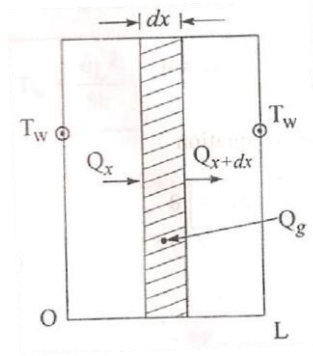
In many practical cases, there is a heat generation within the system. Typical examples are

- Electric coils
- Resistance heater
- Nuclear reactor
- Combustion of fuel in the fuel bed of boiler furnaces.

PLANE WALL WITH INTERNAL HEAT GENERATION

Consider a slab of thickness L, thermal conductivity k, as shown in fig. 1.10.

Consider a small elemental area of thickness δx .



From Fourier's law of conduction, we know that

$$\text{Heat transfer at } x, Q_x = -kA \frac{dT}{dx} \quad \dots\dots (1.51)$$

Heat conducted out at $x + dx$

$$Q_{x+dx} = -kA \frac{dT}{dx} - kA \frac{d^2T}{dx^2} dx \quad \dots\dots (1.52)$$

Heat generated within dx

$$Q_g = qAdx \quad \dots\dots (1.53)$$

We know that,

$$\begin{aligned} Q_x + Q_g &= Q_{x+dx} \\ \implies -kA \frac{dT}{dx} + qAdx &= -kA \frac{dT}{dx} - kA \frac{d^2T}{dx^2} dx \\ \implies qAdx &= -kA \frac{d^2T}{dx^2} \\ \implies kA \frac{d^2T}{dx^2} + qAdx &= 0 \\ \implies \frac{d^2T}{dx^2} + \frac{q}{k} &= 0 \end{aligned} \quad \dots\dots (1.54)$$

Integrating above equation

$$\begin{aligned} \implies \int \frac{d^2T}{dx^2} + \int \frac{q}{k} dx &= 0 \\ (1.54) \quad \implies \frac{dT}{dx} + \frac{q}{k} x &= C_1 \end{aligned} \quad \dots\dots (1.55)$$

Integrating

$$\begin{aligned}
 & \Rightarrow \int \frac{dT}{dx} + \frac{q}{k} \int dx = \int C_1 dx + C_2 \\
 & \Rightarrow T = -\frac{1}{2} \frac{q}{k} x^2 + C_1 x + C_2 \quad \dots\dots\dots (1.56)
 \end{aligned}$$

The temperature on the two faces of the slab (T_w) is the same because it loses the same amount of heat by convection on two sides.

Apply boundary conditions; $C_1 = 0$

$$(1.56) \Rightarrow T = -\frac{1}{2} \frac{q}{k} x^2 + C_2$$

$$\text{Apply } T = T_w, x = \frac{L}{2}$$

$$\begin{aligned}
 \Rightarrow T_w &= -\frac{1}{2} \frac{q}{k} \left(\frac{L}{2}\right)^2 + C_2 \\
 C_2 &= T_w + \frac{1}{2} \frac{q}{k} \left(\frac{L}{2}\right)^2
 \end{aligned}$$

$$C_2 = T_w + \frac{qL^2}{8k}$$

Substituting C_1 and C_2 value in equation (1.56)

$$\begin{aligned}
 T &= -\frac{1}{2} \frac{q}{k} x^2 + 0 + T_w + \frac{qL^2}{8k} \\
 \Rightarrow T &= T_w + \frac{q}{8k} (L^2 - 4x^2) \quad \dots\dots\dots (1.57)
 \end{aligned}$$

The maximum temperature $T_{\max}$ (at the centre) is obtained by putting $x = 0$ in Equation (1.57).

$$\boxed{T_{\max} = T_w + \frac{qL^2}{8k}} \quad \dots\dots\dots (1.58)$$

Heat flow rate

$$Q = \frac{1}{2} qAL$$

Heat transfer by convection

$$\begin{aligned} Q &= hA(T_w - T_\infty) \\ \implies Q &= \frac{1}{2} qAL = hA(T_w - T_\infty) \\ \implies \frac{1}{2} qAL &= hAT_w - hAT_\infty \\ \implies hAT_w &= hAT_\infty + \frac{1}{2} qAL \\ \implies Tw &= T_\infty + \frac{qL}{2h} \end{aligned}$$

Surface or Wall temperature

$$\boxed{Tw = T_\infty + \frac{qL}{2h}} \quad \dots\dots\dots (1.59)$$

CYLINDER WITH INTERNAL HEAT GENERATION

Consider a cylinder of radius r and thermal conductivity k . Heat is generated (Q_g) in the cylinder due to passage of an electric current.

From Fourier's law of conduction, we know that,

$$r \frac{d^2 T}{dr^2} + \frac{q_r}{k} = 0 \quad \dots\dots\dots (1.60)$$

Integrating

$$\begin{aligned}
 r \int \frac{d^2 T}{dr^2} + \int \frac{q_r}{k} &= \int 0 \\
 \implies r \frac{dT}{dr} + \frac{q r^2}{k} &= C_1 \\
 \implies \frac{dT}{dr} + \frac{qr}{2k} &= \frac{C_1}{r}
 \end{aligned}$$

Integrating

$$\begin{aligned}
 \int \frac{dT}{dr} + \int \frac{qr}{2k} &= \int \frac{C_1}{r} \\
 \implies T + \frac{qr^2}{2k \times 2} &= C_1 \ln r + C_2 \quad \dots\dots (1.61)
 \end{aligned}$$

$$\implies T = -\frac{qr^2}{2k} + C_1 \ln r + C_2$$

Apply boundary conditions

$$C_1 = 0$$

$$(1.61) \implies T_w = -\frac{qr_0^2}{2k} + C_2 \quad [\text{Put } T = T_w, r = r_0]$$

$$\implies C_2 = T_w + \frac{qr_0^2}{2k}$$

Apply C_1 and C_2 value in Equation (1.61)

$$\implies T = -\frac{qr^2}{4k} + 0 + T_w + \frac{qr_0^2}{4k}$$

$$T = T_w + \frac{q}{4k} [r_0^2 - r^2]$$

At centre

$$r = 0, T = T_{\max}$$

$$\implies T_{\max} = T_w + \frac{q}{4k} [r_0^2]$$

$$\text{Maximum Temperature, } T_{\max} = T_w + \frac{q}{4k} [r_0^2]$$

..... (1.62)

We know that,

Heat generated

$$Q = \pi r_0^2 L q$$

..... (1.63)

Heat transfer due to convection

$$Q = hA(A_w - T_\infty)$$

$$Q = h \times 2\pi r_0 L (T_w - T_\infty)$$

..... (1.64)

Equation (1.63) and (1.64)

$$\pi r_0^2 L q = h \times 2\pi r_0 L (T_w - T_\infty)$$

$$r_0 q = h \times 2 \times (T_w - T_\infty)$$

$$r_0 q = 2hT_w - 2hT_\infty$$

$$2hT_w = r_0 q + 2hT_\infty$$

$$T_w = T_\infty + \frac{r_0 q}{2h}$$

$$\text{Surface temperature, } T_w = T_\infty + \frac{r_0 q}{2h}$$

..... (1.65)

Similarly,

For sphere, temperature at the centre

$$T_c = T_w + \frac{q r_0^2}{6k}$$

..... (1.66)

$$\Rightarrow \frac{q}{4\pi r^2} = \frac{250}{4/3\pi r^3}$$

$$\Rightarrow q = \frac{250 \times 4 \times \pi (0.050)^2}{4/3\pi \times (0.050)^3}$$

$$\Rightarrow q = 15,000 \text{ W/m}^3$$

Temperature at the centre of the sphere

$$T_c = T_w + \frac{qr^2}{6k} \quad [\text{From Equn no. (1.66)}]$$

$$= 281 + \frac{1555 \times (0.050)^2}{6 \times 0.18}$$

$$T_c = 315.7 \text{ K}$$

Result:

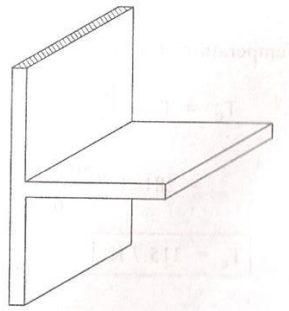
- Heat generated, $q = 15,000 \text{ W/m}^3$
- Center temperature, $T_c = 315.7 \text{ K}$

FINS

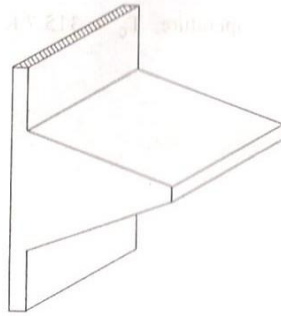
It is possible to increase the heat transfer rate by increasing the surface of heat transfer. The surfaces used for increasing heat transfer are called extended surfaces of fins.

Types of fins

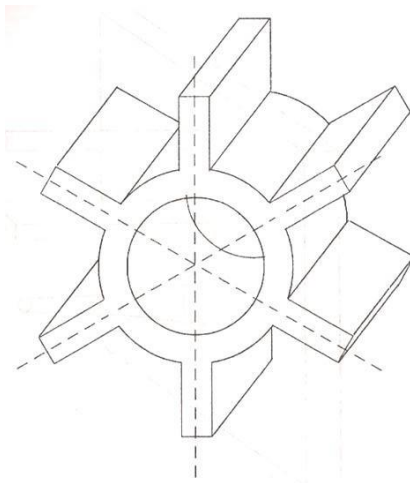
Some common types of fin configuration are shown in fig. 1.11.



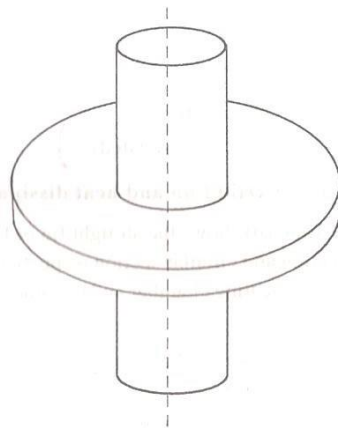
(i) Uniform straight fin



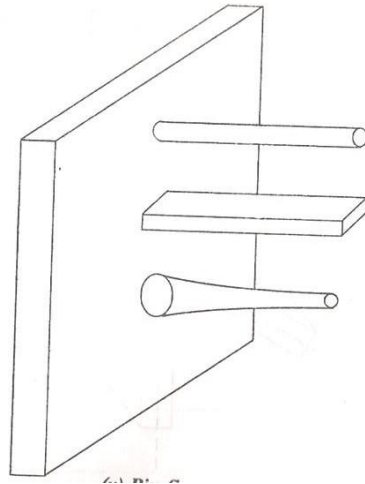
(ii) Tapered straight fin



(iii) Splines



(iv) Annular fin



(v) *Pin fins*

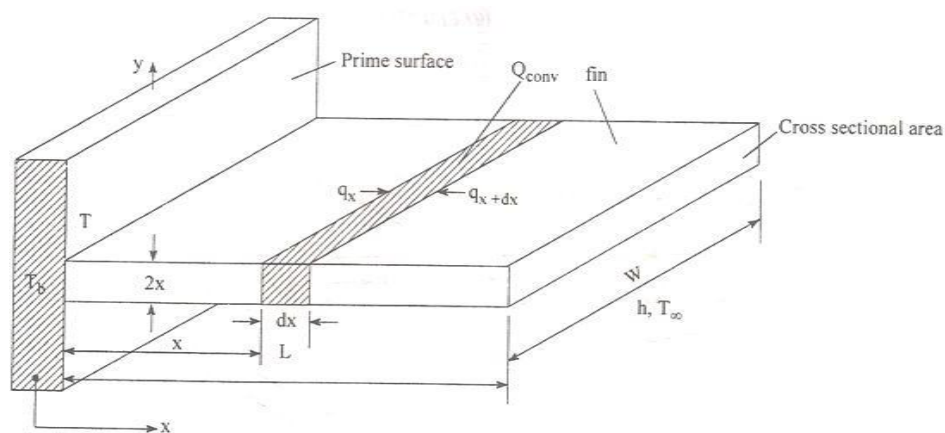
Commonly there are three types of fin

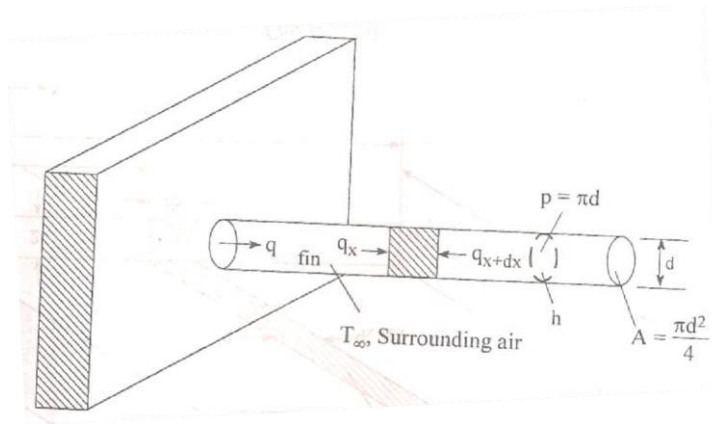
- Infinitely long fin
- Short fin (end is insulated)
- Short fin (end is insulated)

TEMPERATURE DISTRIBUTION AND HEAT DISSIPATION IN FIN

Fig. 1.12 (a) and (b) shows the straight fin or longitudinal fin of rectangular section and circular section respectively. One end of the fin is enclosed in a heating chamber and the other end is exposed to atmospheric air.

Heat is transferred across the rectangular fin and circular rod by conduction. From the surface of the fin, heat is transferred to air by convection. Let us consider a small elemental area of thickness dx , which is at a distance of x from the base.





A steady state conditions, heat balance equation for that element is as follows.

Heat conducted into the element = Heat conducted out of the element + heat convected to the surrounding air.

$$\Rightarrow Q_x = Q_{x+dx} + Q_{conv} \quad \dots\dots (1.67)$$

Where,

$$Q_x = -kA \left[\frac{dT}{dx} \right]$$

$$Q_{x+dx} = -kA \left[\frac{dT}{dx} \right]_{x+dx} = -kA \left[\frac{dT}{dx} + \frac{d^2T}{dx^2} dx \right]$$

$$Q_{conv} = hA(T - T_\infty)$$

$$= h(pdx)(T - T_\infty)$$

Substituting Q_x , Q_{x+dx} and Q_{conv} values in equation (1.67)

(1.67)

$$\Rightarrow -kA \frac{dT}{dx} = -kA \left[\frac{dT}{dx} + \frac{d^2T}{dx^2} dx \right] + h(pdx)(T - T_\infty)$$

$$\Rightarrow \frac{d^2T}{dx^2} = \frac{hP}{kA}(T - T_\infty)$$

$$\Rightarrow \frac{d^2T}{dx^2} = \frac{hP}{kA}(T - T_\infty)$$

$$\Rightarrow \frac{d^2T}{dx^2} = \frac{hP}{kA}(T - T_\infty)$$

$$\Rightarrow \frac{d^2T}{dx^2} = \frac{hP}{kA}(T - T_\infty)$$

$$\text{Where, } m^2 = \frac{hP}{kA}$$

$$\begin{aligned} \Rightarrow m &= \sqrt{\frac{hP}{kA}} \\ \Rightarrow \frac{d^2 T}{dx^2} - m^2 \theta &= 0 \end{aligned} \quad \dots\dots (1.68)$$

$$[\because \theta = T - T_\alpha]$$

Equation (1.68) shows that the temperature is a function of x and m . It is a second order, linear differential equation. Its general solution is,

$$\theta = C_1 e^{-mx} + C_2 e^{mx} \quad \dots\dots (1.69)$$

The temperature distribution and heat dissipation depends upon the following fin conditions.

Case (i): Infinitely long fin

If a fin is infinitely long, the temperature at its end is equal to that of the surrounding fluid.

$$\text{At } x=0; T = T_b \text{ and } \text{At } X = \infty; T = T_\alpha$$

T_b – Base temperature of fin

From equation (1.69), we know that,

$$\begin{aligned} \theta &= C_1 e^{-mx} + C_2 e^{mx} \\ T - T_\alpha &= C_1 e^{-mx} + C_2 e^{mx} \end{aligned} \quad \dots\dots (1.70)$$

$$[\because \theta = T - T_\alpha]$$

Substituting

$$\text{At } X = 0; T = T_b$$

$$(1.70) \Rightarrow T_b - T_\alpha = C_1 e^0 + C_2 e^0$$

$$\Rightarrow \boxed{T_b - T_\alpha = C_1 + C_2} \quad \dots\dots (1.71)$$

Substituting

At $X = \alpha; T = T_\alpha$ in equation (1.70)

$$\implies (T_\alpha - T_\alpha) = C_1 e^{-m\alpha} + C_2 e^{m\alpha}$$

$$\implies 0 = C_1 e^{-m\alpha} + C_2 e^{m\alpha}$$

$$\implies C_2 e^{m\alpha} = 0$$

$$e^{m\alpha} \neq 0, \text{ So, } \boxed{C_2 = 0}$$

Substituting $C_2 = 0$ value in equation (1.71)

$$(1.70) \implies T_b - T_\alpha = C_1 + 0$$

$$\boxed{\implies T_b - T_\alpha = C_1}$$

Substituting C_1 and C_2 value in equation (1.70)

$$(1.70) \implies T - T_\alpha = (T_b - T_\alpha) e^{-mx} + 0$$

$$\implies \frac{T - T_\alpha}{T_b - T_\alpha} = e^{-mx}$$

$$\boxed{\text{Temperature distribution of fin, } \frac{T - T_\alpha}{T_b - T_\alpha} = e^{-mx}}$$

..... (1.72)

Where,

T_b – Base temperature, K

T_α - Surrounding temperature, K

T – Intermediate temperature, K

x - Distance, m

$$m = \sqrt{\frac{hP}{kA}}$$

h – heat transfer co-efficient, W/m²K

P – Perimeter, m

k – Thermal conductivity, W/mK

A – Area, m²

Heat dissipation through the fin is obtained by integrating the heat lost by convection over the entire fin surface.

We know that,

$$\text{Heat lost by convection, } Q_{conv} = hA(T - T_{\alpha})$$

$$\implies Q = hPdx(T - T_{\alpha})$$

$$\implies Q = \int_0^{\alpha} hP(T - T_{\alpha})dx$$

$$\implies Q = \int_0^{\alpha} hPe^{-mx}(T_b - T_{\alpha})dx$$

$$\left[\because \frac{T - T_{\alpha}}{T_b - T_{\alpha}} = e^{-mx} \implies T - T_{\alpha} = (T_b - T_{\alpha})e^{-mx} \right]$$

$$\implies Q = hP(T_b - T_{\alpha}) \int_0^{\alpha} e^{-mx} dx$$

$$= hP(T_b - T_{\alpha}) - \frac{1}{m} [e^{-mx}]_0^{\alpha}$$

$$= hP(T_b - T_{\alpha}) \times -\frac{1}{m} (-1)$$

$$= hP(T_b - T_{\alpha}) \times \frac{1}{m}$$

$$= \frac{1}{m} \times hP \times (T_b - T_{\alpha})$$

$$= \frac{1}{\sqrt{\frac{hP}{kA}}} \times hP \times (T_b - T_{\alpha}) \left[\because m = \sqrt{\frac{hP}{kA}} \right]$$

$$Q = \sqrt{hPkA}(T_b - T_{\alpha})$$

Heat transferred, $Q = \sqrt{hPkA}(T_b - T_{\alpha})$

.... (1.73)

Case (ii): Fin with insulated end (Short fin)

The fin has a finite length and the tip of fin is insulated.

$$\begin{aligned} \text{At } x = L; \frac{dT}{dx} &= 0; \\ \text{At } x = 0; T &= T_b \end{aligned}$$

From equation (1.70), we know that,

$$\begin{aligned} (T - T_\alpha) &= C_1 e^{-mx} + C_2 e^{mx} \\ \Rightarrow \frac{dT}{dx} &= C_1 e^{-mx} \times (-m) + C_2 e^{mx} \times m \end{aligned}$$

Applying the first boundary condition, i.e., at $x = L$, $\frac{dT}{dx} = 0$

$$\begin{aligned} \Rightarrow 0 &= C_1 e^{-mL} \times -m + C_2 e^{mL} \times m \\ \Rightarrow mC_1 e^{-mL} &= mC_2 e^{mL} \\ C_1 e^{-mL} &= C_2 e^{mL} \end{aligned}$$

$$\boxed{C_1 e^{-mL} = C_2 e^{mL}} \quad \dots\dots (1.74)$$

From equation (1.70), we know that,

$$(T - T_\alpha) = C_1 e^{-mx} + C_2 e^{mx}$$

Applying the Second boundary condition, i.e., at $x = 0$, $T = T_b$

$$\begin{aligned} \Rightarrow (T_b - T_\alpha) &= C_1 e_0 + C_2 e_0 \\ \Rightarrow (T_b - T_\alpha) &= C_1 + C_2 \\ \Rightarrow T_b - T_\alpha &= C_2 e^{2mL} + C_2 \quad \left[\because C_1 = C_2 e^{2mL} \right] \\ \Rightarrow T_b - T_\alpha &= C_2 [e^{2mL} + 1] \end{aligned}$$

$$\boxed{C_2 = \frac{T_b - T_\alpha}{[e^{2mL} + 1]}} \quad \dots (1.75)$$

Substituting C_2 value in equation (1.74)

$$\begin{aligned} \Rightarrow C_1 &= \left[\frac{T_b - T_\alpha}{e^{2mL} + 1} \right] \times e^{2mL} \\ \Rightarrow C_1 &= \frac{T_b - T_\alpha}{[e^{2mL} + 1]} \times \frac{1}{e^{-2mL}} \\ \Rightarrow C_1 &= \frac{T_b - T_\alpha}{e^{2mL} \times e^{-2mL} + e^{-2mL}} \end{aligned}$$

$$\Rightarrow \boxed{C_1 \frac{T_b - T_\alpha}{1 + e^{-2mL}}} \quad \dots (1.76)$$

Substituting C_1 in equation (1.70)

$$\begin{aligned} \Rightarrow (T - T_\alpha) &= \frac{(T_b - T_\alpha)}{1 + e^{-2mL}} \times e^{-mx} + \frac{(T_b - T_\alpha)}{1 + e^{2mL}} \times e^{mx} \\ \Rightarrow T - T_\alpha &= (T_b - T_\alpha) \left[\frac{e^{-mx}}{1 + e^{-2mL}} + \frac{e^{mx}}{1 + e^{2mL}} \right] \\ \Rightarrow \frac{(T - T_\alpha)}{(T_b - T_\alpha)} &= \left[\frac{e^{-mx}}{1 + e^{-2mL}} + \frac{e^{mx}}{1 + e^{2mL}} \right] \end{aligned}$$

Multiplying the numerator and denominator by e^{mL} and e^{-mL}

$$\begin{aligned} \Rightarrow \frac{(T - T_\alpha)}{(T_b - T_\alpha)} &= \frac{e^{-mx}}{1 + e^{-2mL}} \times \frac{e^{mL}}{e^{mL}} + \frac{e^{mx}}{1 + e^{2mL}} \times \frac{e^{-mL}}{e^{-mL}} \\ &= \frac{e^{-m(x-L)}}{e^{mL} + e^{-mL}} + \frac{e^{m(L-x)}}{e^{-mL} + e^{mL}} \\ \Rightarrow \frac{T - T_\alpha}{T_b - T_\alpha} &= \frac{e^{m(L-x)} + e^{-m(L-x)}}{e^{mL} + e^{-mL}} \quad \dots\dots (1.77) \end{aligned}$$

In terms of hyperbolic function it can be written as,

$$\frac{T - T_\alpha}{T_b - T_\alpha} = \frac{\cosh m(L-x)}{\cosh mL}$$

Temperature distribution of fin with insulated end

$$\boxed{\frac{T - T_\alpha}{T_b - T_\alpha} = \frac{\cosh m(L-x)}{\cosh mL}} \quad \dots\dots\dots (1.78)$$

$$\begin{aligned} \Rightarrow T - T_\alpha &= (T_b - T_\alpha) \frac{\cosh m(L-x)}{\cosh mL} \\ \Rightarrow \frac{dT}{dx} &= (T_b - T_\alpha) \times -m \frac{\sinh m(L-x)}{\cosh mL} \end{aligned}$$

We know that,

$$\text{Heat transferred, } Q = -kA \frac{dT}{dx}$$

$$= -kA(T_b - T_\alpha) \times -m \times \frac{\sinh m(L-x)}{\cosh mL}$$

$$Q = kAm(T_b - T_\alpha) \times \frac{\sinh m(L-x)}{\cosh mL}$$

$$\text{at } x = 0$$

$$\Rightarrow Q = kAm(T_b - T_\alpha) \times \frac{\sinh(mL)}{\cosh(mL)}$$

$$= kAm(T_b - T_\alpha) \tanh(mL)$$

$$= kA \times \sqrt{\frac{hP}{kA}} (T_b - T_\alpha) \tanh(mL)$$

$$\left[\begin{array}{l} \text{[} hP \text{]} \\ \therefore m = \sqrt{\frac{hP}{kA}} \end{array} \right]$$

$$= \sqrt{hPkA} (T_b - T_\alpha) \tanh(mL)$$

Heat transferred for insulated fin $Q = \sqrt{hPkA} (T_b - T_\alpha) \tanh(mL)$

.....(1.79)

Applications

The main application of fins are

- Cooling of electronic components
- Cooling of motor cycle engines.
- Cooling of small capacity compressors
- Cooling of transformers
- Cooling of radiators and refrigerators etc.

Fin efficiency

The efficiency of a fin is defined as the ratio of actual heat transferred fin to the maximum possible heat transferred by the fin

$$\mu_{fin} = \frac{Q_{fin}}{Q_{max}}$$

For insulated end

$$\mu_{fin} = \frac{\tanh(mL)}{mL}$$

Fin effectiveness

It is defined as the ratio of heat transfer with fin to heat transfer without fin

$$\text{Fin effectiveness, } E = \frac{Q_{withfin}}{Q_{withoutfin}}$$

For insulated end

$$\text{Fin effectiveness, } E = \frac{\tanh(mL)}{\sqrt{\frac{hA}{kP}}}$$

TRANSIENT HEAT CONDUCTION (OR) UNSTEADY STATE CONDUCTION

If the temperature of a body does not vary with time, it is said to be in a steady state. But if there is an abrupt change in its surface temperature, it attains a steady state after some period. During this period the temperature varies with time and the body is said to be in an unsteady or transient state.

Transient heat conduction occur in cooling of IC engines, automobile engines, boiler tubes, heating and cooling of metal billets, rocket nozzles, electric irons etc.

Transient heat conduction can be divided in to periodic heat flow and non periodic heat flow.

(i) Periodic heat flow

In periodic heat flow, the temperature varies on a regular basis.

Examples: Cylinder of an IC engine,

Surface of earth during a period of 24 hours.

(ii) Non periodic heat flow:

In non periodic heat flow, the temperature at any point within the system varies non-linearly with time.

Examples: heating of an ingot in a furnace, cooling of bars.

Biot Number

The ratio of internal conduction resistance to the surface convection resistance is known as Biot number.

$$\text{Biot Number} = \frac{\text{Internal conduction resistance}}{\text{Surface convection resistance}}$$

$$B = \frac{hL_c}{k}$$

Where

K – Thermal conductivity, W/mK

H – Heat transfer co-efficient, W/m²K

L_c – Characteristic length or significant length

$$\text{Characteristic length, } L_c = \frac{\text{Volume}}{\text{Surface Area}} = \frac{V}{A}$$

For slab:

$$\text{Characteristic length, } L_c = \frac{V}{A} = \frac{A \times L}{2A}$$

$$L_c = \frac{L}{2}$$

Where

L – Thickness of the slab

For cylinder:

$$\text{Characteristic length, } L_c = \frac{R}{2} = \frac{\pi R^2 L}{2\pi RL}$$

$$L_c = \frac{R}{2}$$

Where

R – Radius of cylinder

For sphere:

$$\text{Characteristic length, } L_c = \frac{V}{A}$$

$$\begin{aligned} & \frac{\frac{4}{3}\pi R^3}{4\pi R^2} \\ &= \frac{R}{3} \end{aligned}$$

$$L_c = \frac{R}{3}$$

Where

R – Radius of the sphere.

For Cube:

$$\text{Characteristic length, } L_c = \frac{V}{A}$$

$$= \frac{L^3}{6L^2}$$

$$L_c = \frac{L}{6}$$

Where

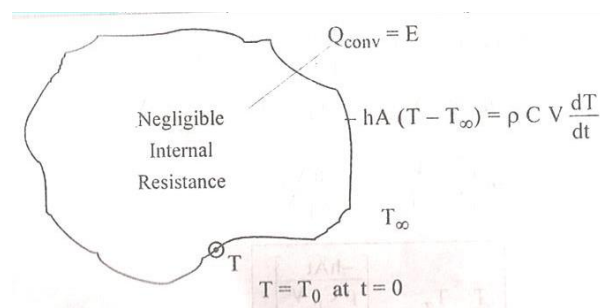
L – Thickness of the cube.

LUMPED HEAT ANALYSIS [NEGLECTIBLE INTERNAL RESISTANCE]

The process in which the internal resistance is assumed as negligible in comparison with its surface resistance is known as Newtonian heating or cooling process.

In a Newtonian heating or cooling process the temperature is considered to be uniform at a given time. Such an analysis is called lumped parameter analysis.

Let us consider a solid whose initial temperature is T^0 and it is placed suddenly in ambient air or any liquid at a constant temperature T_∞ . The transient response of the body can be determined by relating its rate of change of internal energy with convective exchange at the surface.



Convective heat loss from the body = Rate of change of internal energy

$$\Rightarrow -hA(T - T_\infty) = \rho \times C_p \times V \times \frac{dT}{dt}$$

$$\Rightarrow \frac{dT}{T - T_\infty} = \frac{-hA}{\rho C_p V} \times dt$$

Integrating

$$\Rightarrow \int \frac{dT}{T - T_\infty} = \frac{-hA}{\rho C_p V} \times \int dt$$

$$\Rightarrow \ln[T - T_\infty] = \frac{-hA}{\rho C_p V} t + C_1 \quad \dots (1.80)$$

Apply boundary conditions.

At $t = 0$, $T = T_0$

$$(1.80) \Rightarrow \ln[T_0 - T_\infty] = 0 + C_1$$

$$C_1 = \ln[T_0 - T_\infty]$$

Substituting C_1 value in equation (1.80)

$$\Rightarrow \ln\left[\frac{T - T_\infty}{T_0 - T_\infty}\right] = \frac{-hA}{\rho C_p V} t + \ln\left[\frac{T_0 - T_\infty}{T_0 - T_\infty}\right]$$

$$\Rightarrow \ln\left[\frac{T - T_\infty}{T_0 - T_\infty}\right] - \ln\left[\frac{T_0 - T_\infty}{T_0 - T_\infty}\right] = \frac{-hA}{\rho C_p V} t$$

$$\Rightarrow \ln\left[\frac{T - T_\infty}{T_0 - T_\infty}\right] = \frac{-hA}{\rho C_p V} t$$

$$\Rightarrow \frac{T - T_\infty}{T_0 - T_\infty} = e^{\left[\frac{-hAt}{\rho C_p V}\right]} \quad \dots\dots (1.81)$$

Where

T_0 – Initial temperature of the solid, K

T – Intermediate temperature of the solid, K

T_∞ - Surface temperature of the solid (or) Final temperature of the solid, K

h – Heat transfer co-efficient, W/m²K

A – Surface area of body, m²

ρ – Density of the body, kg/m³

V – Volume of the body, m³

C_p – Specific heat of the body, J/kg K.

t – Time, s.

Note

1. In lumped parameter system, Biot number value is less than 0.1.

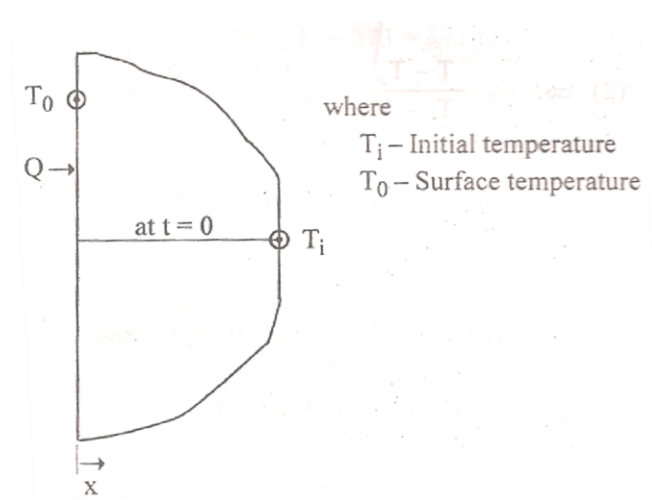
$$\text{i.e., } B_i < 0.1 \Rightarrow \frac{hL_c}{k} < 0.1$$

2. T_0 – Initial temperature, K
 T – Intermediate temperature, K
 T_{∞} - Surface temperature or Final temperature, K

HEAT FLOW IN SEMI-INFINITE SOLIDS

A solid which extends itself infinitely in all directions of space is known as infinite solid. If an infinite solid is split in the middle by a plane, each half is known as semi infinite solid.

In a semi infinite solid, at any instant of time, there is always a point where the effect of heating (or cooling) at one of its boundaries is not felt at all. At this point the temperature remains unchanged.



Semi Infinite Plate

Consider a semi infinite body and it extends to infinity in the +ve x direction. The entire body is initially at uniform temperature T_i including the surface at $x = 0$ is suddenly raised to T_0 .

The governing equation is

$$\frac{d^2 T}{dx^2} = \frac{1}{\alpha} \frac{dT}{dt}$$

The boundary conditions are

- $T(x, 0) = T_i$
- $T(0, t) = T_0$ for $t > 0$
- $T(\infty, t) = T_i$ for $t > 0$

The analytical solution for this case is given by

$$\frac{T_x - T_0}{T_i - T_0} = \text{erf} \left[\frac{x}{2\sqrt{\alpha t}} \right] \quad \dots (1.82)$$

Where erf indicates “error function of” and the definition of error function is generally available in mathematical texts. Usually tabulation of error values are available in data books.

α - Thermal diffusivity, m²/s

t – Time, s

X – Distance, m

T_i – Initial temperature, K

T₀ – Surface temperature (or) Final temperature, K

T_x - Intermediate temperature, K

Note

1. In semi infinite solid, heat transfer co-efficient or biot number value is ∞ .

i.e., $h \Rightarrow \infty$

or

$B_i \Rightarrow \infty$

2. T_i – Initial temperature, K

T₀ – Surface temperature (or) Final temperature, K

T_x - Intermediate temperature, K

TRANSIENT HEAT FLOW IN AN INFINITE PLATE

A solid which extends itself infinitely in all directions of space is known as infinite solid.

Consider an infinite flat plate of uniform thickness 2L as shown in fig. 1.15, which is initially at a uniform temperature of T_i. It is suddenly exposed to a large mass of fluid having a temperature T_∞. This temperature is assumed to be constant throughout the process of cooling or heating. The plate is extended to infinity in the y and z directions.

The heat transfer co-efficient between the surface of the plate and the fluid on both sides is assumed to be constant. The center of the plate is selected as the origin.

The governing differential equation is

$$\frac{d^2 T}{dx^2} = \frac{1}{\alpha} \frac{dT}{dt}$$

The boundary conditions are

- $T|_{t=0} = T_0 = T_i$
- $T|_{x=0} = 0, \frac{dT}{dx} = 0$
- $T|_{x=L} = T_\infty, kA \frac{dT}{dx} = hA[T_0 - T_\infty]$

The solution of the above differential equation with this boundary condition is given by

$$\frac{T_0 - T_\infty}{T_i - T_\infty} = f\left[\frac{x}{L}, \frac{hL}{k}, \frac{\alpha t}{L^2}\right]$$

From this equation, we know that conduction resistance is not negligible. The temperature history becomes a function of biot number $\left[\frac{hL_c}{k}\right]$, Fourier number $\left[\frac{\alpha t}{L^2}\right]$ and the

dimensionless parameter $\left[\frac{x}{L}\right]$ which indicates the location of point within the plate where

temperature is to be obtained. The dimensionless parameter $\left[\frac{x}{L}\right]$ is replaced by $\left[\frac{r}{R}\right]$ in case of cylinders and spheres.

Heisler has prepared charts for graphical solutions of the unsteady state conduction problems. These charts have been constructed in non-dimensional parameters. The charts are suitable for problems with a finite surface and internal resistance. For such case the biot number lies between 0 and 100.

These heiler charts were further extended and improved by grobe.

The heiler and grober charts are used to solve the problems of sudden immersion of plate, cylinder or sphere into a fluid.

Note: For infinite solids,

Take T_i – Initial temperature – K

T_∞ - Final temperature – K

T_0 - Center line temperature – K

T_x - Intermediate temperature – K

The infinite solids, biot number value is in between 0.1 and 100 i.e., $0.1 < B_i < 100$.

1. The wall of a cold room is composed of three layer. The outer layer is brick 30 cm thick. The middle layer is cork 20 cm thick, the inside layer is cement 15 cm thick. Temperatures of the outside air is 25°C and on the inside air is -20°C . The film co-efficient for outside air and brick is $55.4\text{ W/m}^2\text{ K}$. Film co-efficient for inside air and cement is $17\text{ W/m}^2\text{ K}$. Find heat flow rate.

Take

$$k \text{ for brick} = 2.5\text{ W/mK}$$

$$k \text{ for cork} = 0.05\text{ W/mK}$$

$$k \text{ for cement} = 0.28\text{ W/mK}$$

Given:

$$\text{Thickness of brick, } L_3 = 30\text{ cm} = 0.3\text{ m}$$

$$\text{Thickness of cork, } L_2 = 20\text{ cm} = 0.2\text{ m}$$

$$\text{Thickness of cement, } L_1 = 15\text{ cm} = 0.15\text{ m}$$

$$\text{Inside air temperature, } T_a = -20^{\circ}\text{C} + 273 = 253\text{ K}$$

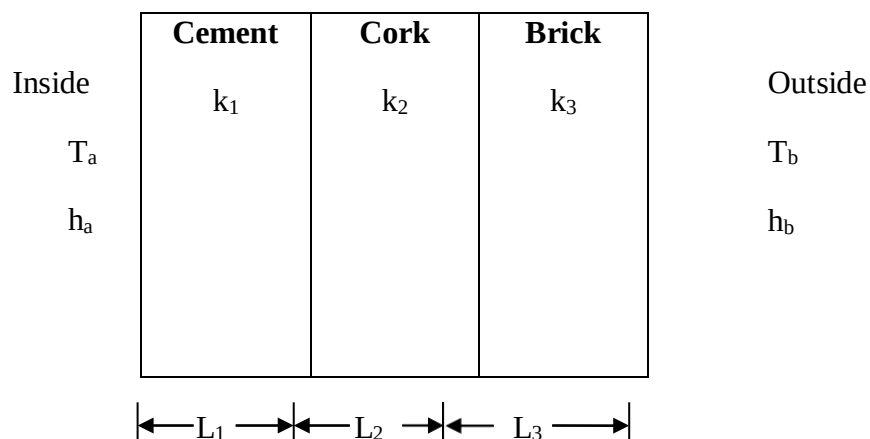
$$\text{Outside air temperature, } T_b = 25^{\circ}\text{C} + 273 = 293\text{ K}$$

$$\text{Film co-efficient of r inner side, } h_a = 17\text{ W/m}^2\text{ K}$$

$$k_{\text{brick}} = k_3 = 2.5\text{ W/mK}$$

$$k_{\text{cork}} = k_2 = 0.05\text{ W/mK}$$

$$k_{\text{cement}} = k_1 = 0.28\text{ W/mK}$$



To find:

Heat flow rate (Q/A)

Solution:

Heat flow through composite wall is given by

$$Q = \frac{\Delta T_{\text{overall}}}{R} \quad [\text{From Equn no. 1.42 or HMT Date book page No. 43 and 44}]$$

Where

$$\begin{aligned}\Delta T &= T_a - T_b \\ R &= \frac{1}{h_a A} + \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \frac{L_3}{k_3 A} + \frac{1}{h_b A} \\ \Rightarrow Q &= \frac{[T_a - T_b]}{\frac{1}{h_a A} + \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \frac{L_3}{k_3 A} + \frac{1}{h_b A}} \\ \Rightarrow Q / A &= \frac{[T_a - T_b]}{\frac{1}{h_a} + \frac{L_1}{k_1} + \frac{L_2}{k_2} + \frac{L_3}{k_3} + \frac{1}{h_b}} \\ \Rightarrow Q / A &= \frac{253 - 298}{\frac{1}{17} + \frac{0.15}{0.28} + \frac{0.2}{0.05} + \frac{0.3}{2.5} + \frac{1}{55.4}}\end{aligned}$$

$$Q/A = -9.5 \text{ W/m}^2$$

The negative sign indicates that the heat flows from the outside into the cold room.

Result:

Heat flow rate, $Q/A = -9.5 \text{ W/m}^2$

2. A wall of a cold room is composed of three layer. The outer layer is brick 20 cm thick, the middle layer is cork 10 cm thick, the inside layer is cement 5 cm thick. The temperature of the outside air is 25°C and that on the inside air is -20°C . The film co-efficient for outside air and brick is $45.4 \text{ W/m}^2\text{K}$ and for inside air and cement is $17 \text{ W/m}^2\text{K}$.

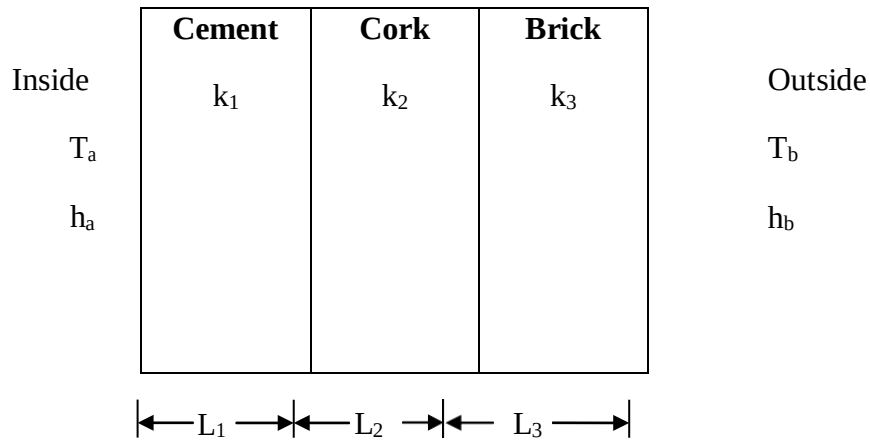
Take

k for brick = 3.45 W/mK

k for cork = 0.043 W/mK

k for cement = 0.294 W/mK

Given:



Thickness of brick, $L_3 = 20 \text{ cm} = 0.2 \text{ m}$

Thickness of cork, $L_2 = 10 \text{ cm} = 0.1 \text{ m}$

Thickness of cement, $L_1 = 5 \text{ cm} = 0.05 \text{ m}$

Outside air temperature, $T_b = 25^\circ \text{ C} + 273 = 293 \text{ K}$

Inside air temperature, $T_a = -20^\circ \text{ C} + 273 = 253 \text{ K}$

Film co-efficient for outside air and brick, $h_b = 45.4 \text{ W/m}^2\text{K}$

Film co-efficient for inside air and cement, $h_a = 17 \text{ W/m}^2\text{K}$

$k_3 = 3.45 \text{ W/mK}$

$k_2 = 0.043 \text{ W/mK}$

$k_1 = 0.294 \text{ W/mK}$

To find:

1. Heat flow rate
2. Thermal resistance of the wall

Solution:

Heat flow through composite wall is given by

$$Q = \frac{\Delta_{\text{Overall}}}{R} \quad [\text{From Equn no. (1.42) (or) HMT Data book page No. 43 and 44}]$$

Where

$$\begin{aligned} \Delta T &= T_a - T_b \\ R &= \frac{1}{h_a A} + \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \frac{L_3}{k_3 A} + \frac{1}{h_b A} \\ \Rightarrow Q &= \frac{[T_a - T_b]}{\frac{1}{h_a A} + \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \frac{L_3}{k_3 A} + \frac{1}{h_b A}} \\ \Rightarrow Q/A &= \frac{[T_a - T_b]}{\frac{1}{h_a} + \frac{L_1}{k_1} + \frac{L_2}{k_2} + \frac{L_3}{k_3} + \frac{1}{h_b}} \\ \Rightarrow Q/A &= \frac{253 - 298}{\frac{1}{17} + \frac{0.05}{0.294} + \frac{0.1}{0.043} + \frac{0.2}{3.45} + \frac{1}{45.4}} \end{aligned}$$

$$Q/A = -17.081 \text{ W/m}^2$$

The negative sign indicates that the heat flows from the outside into the cold room.

Thermal Resistance

$$R = \frac{1}{h_a A} + \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \frac{L_3}{k_3 A} + \frac{1}{h_b A}$$

For Unit Area

$$\Rightarrow R = \frac{1}{h_a} + \frac{L_1}{k_1} + \frac{L_2}{k_2} + \frac{L_3}{k_3} + \frac{1}{h_b}$$

$$\Rightarrow R = \frac{1}{17} + \frac{0.05}{0.294} + \frac{0.1}{0.043} + \frac{0.2}{3.45} + \frac{1}{45.4}$$

$$\Rightarrow R = 2.634 \text{ K / W}$$

Result:

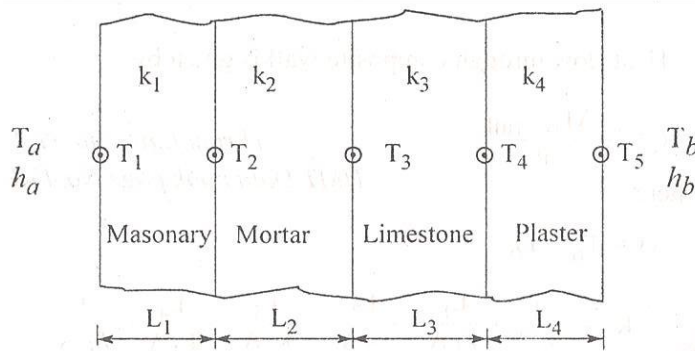
1. Heat flow rate, $Q/A = -17.081 \text{ W/m}^2$
2. Thermal resistance, $R = 2.634 \text{ K/W}$

3. A wall is constructed of several layers. The first layer consists of masonry brick 20 cm thick of thermal conductivity 0.66 W/mK, the second layer consists of 3 cm thick mortar of thermal conductivity 0.6 W/mK, the third layer consists of 8 cm thick lime stone of thermal conductivity 0.58 W/mK and the outer layer consists of 1.2 cm thick plaster of thermal conductivity 0.6 W/mK. The heat transfer co-efficient on the interior and exterior of the wall are 5.6 W/m²K and 11 W/m² K respectively. Interior room temperature is 22° C and outside air temperature is -5° C.

Calculate

- Overall heat transfer co-efficient
- Overall thermal resistance
- The rate of heat transfer
- The temperature at the junction between the mortar and the limestone

Given:



Thickness of masonry, $L_1 = 20 \text{ cm} = 0.20 \text{ m}$

Thermal conductivity, $k_1 = 0.66 \text{ W/mK}$

Thickness of mortar, $L_2 = 3 \text{ cm} = 0.03 \text{ m}$

Thermal conductivity of mortar, $k_2 = 0.6 \text{ W/mK}$

Thickness of limestone, $L_3 = 8 \text{ cm} = 0.08 \text{ m}$

Thermal conductivity, $k_3 = 0.58 \text{ W/mK}$

Thickness of Plaster, $L_4 = 1.2 \text{ cm} = 0.012 \text{ m}$

Thermal conductivity, $k_4 = 0.6 \text{ W/mK}$

Interior heat transfer co-efficient $h_a = 5.6 \text{ W/m}^2\text{K}$

Exterior heat transfer co-efficient $h_b = 11 \text{ W/m}^2\text{K}$

Inside air temperature, $T_a = 22^\circ\text{C} + 273 = 295 \text{ K}$

Outside air temperature, $T_b = -5^\circ\text{C} + 273 = 268 \text{ K}$

To find:

- a) Overall heat transfer co-efficient, U
- b) Overall thermal resistance, (R)
- c) Heat transfer/ m^2 , (Q/A)
- d) The temperature at the junction between the Mortar and the limestone, (T_3)

Solution:

Heat flow through composite wall is given by

$$Q = \frac{\Delta T_{\text{Overall}}}{R} \quad [\text{From Equn no. (1.42) or HMT Data book page No. 43 \& 44}]$$

Where

$$\Delta T = T_a - T_b$$

$$R = \frac{1}{h_a A} + \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \frac{L_3}{k_3 A} + \frac{1}{h_b A}$$

$$\Rightarrow Q = \frac{T_a - T_b}{\frac{1}{h_a A} + \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \frac{L_3}{k_3 A} + \frac{1}{h_b A}}$$

$$\Rightarrow Q/A = \frac{295 - 268}{\frac{1}{5.6} + \frac{0.20}{0.66} + \frac{0.03}{0.6} + \frac{0.08}{0.58} + \frac{0.012}{0.6} + \frac{1}{11}}$$

Heat transfer per unit area, $Q/A = 34.56 \text{ W/m}^2$
--

We know that,

$$\text{Heat transfer, } Q = UA(T_a - T_b) \text{ [From equation no. 1.43]}$$

Where, U – overall heat transfer co-efficient

$$\Rightarrow U = \frac{Q}{A \times (T_a - T_b)}$$

$$\Rightarrow U = \frac{34.56}{295 - 268}$$

Overall heat transfer co-efficient, $U = 1.28 \text{ W/m}^2\text{K}$
--

We know that,

Overall thermal resistance (R)

$$R = \frac{1}{h_a A} + \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \frac{L_3}{k_3 A} + \frac{L_4}{k_4 A} + \frac{1}{h_b A}$$

For unit Area

$$R = \frac{1}{h_a} + \frac{L_1}{k_1} + \frac{L_2}{k_2} + \frac{L_3}{k_3} + \frac{L_4}{k_4} + \frac{1}{h_b}$$

$$\frac{1}{5.6} + \frac{0.20}{0.66} + \frac{0.03}{0.6} + \frac{0.08}{0.58} + \frac{0.012}{0.6} + \frac{1}{11}$$

$$R = 0.78 \text{ K/W}$$

Interface temperature between mortasr and the limestorn, T3

Interface temperatures relation

$$Q = \frac{T_a - T_1}{R_a} = \frac{T_1 - T_2}{R_1} = \frac{T_2 - T_3}{R_2} = \frac{T_3 - T_4}{R_3} = \frac{T_4 - T_5}{R_4} = \frac{T_5 - T_b}{R_b}$$

$$\Rightarrow Q = \frac{T_a - T_1}{R_a}$$

$$Q = \frac{295 - T_1}{\frac{1}{h_a A}} \quad \left[\because R_a = \frac{1}{h_a A} \right]$$

$$Q / A = \frac{295 - T_1}{\frac{1}{h_a}}$$

$$34.56 = \frac{295 - T_1}{1 / 5.6}$$

$$T_1 = 288.8 \text{ K}$$

$$\Rightarrow Q = \frac{T_1 - T_2}{R_1}$$

$$Q = \frac{288.8 - T_2}{\frac{L_1}{k_1 A}} \quad \left[\because R_1 = \frac{L_1}{k_1 A} \right]$$

$$Q / A = \frac{288.8 - T_2}{\frac{L_1}{k_1}}$$

$$34.56 = \frac{288.8 - T_2}{\frac{0.20}{0.66}}$$

$$T_2 = 278.3 \text{ K}$$

$$\Rightarrow Q = \frac{T_2 - T_3}{R_2}$$

$$Q = \frac{278.3 - T_3}{\frac{L_2}{k_2 A}} \quad \left[\because R_2 = \frac{L_2}{k_2 A} \right]$$

$$Q / A = \frac{278.3 - T_3}{\frac{L_2}{k_2}}$$

$$34.56 = \frac{278.3 - T_3}{\frac{0.03}{0.6}}$$

$$T_3 = 276.5 \text{ K}$$

Temperature between Mortar and limestone (T_3) is 276.5 K

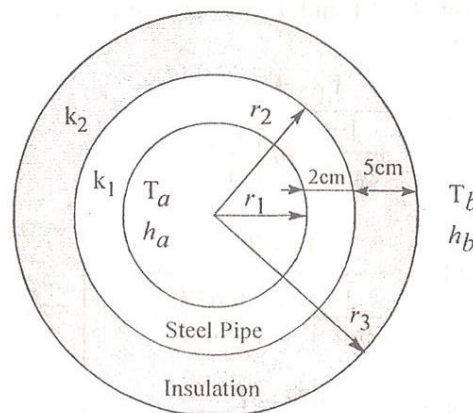
Result:

- Overall heat transfer co-efficient, $U = 1.28 \text{ W/m}^2\text{K}$
- Overall thermal resistance, $R = 0.78 \text{ K/W}$
- Heat transfer, $Q/A = 34.56 \text{ W/m}^2$
- Temperature between mortar and limestone, (T_3) = 276.5 K

4. An insulated steel pipe carrying a hot liquid. Inner diameter of the pipe is 25 cm, wall thickness is 2 cm, thickness of insulation is 5 cm, temperature of hot liquid is 100°C , temperature of surrounding is 20°C , inside heat transfer co-efficient is $730 \text{ W/m}^2\text{K}$ and outside heat transfer co-efficient is $12 \text{ W/m}^2\text{K}$. Calculate the heat loss per meter length of the pipe.

Take $k_{\text{steel}} = 55 \text{ W/mK}$, $k_{\text{insulating material}} = 0.22 \text{ W/mK}$

Given:



Inner diameter, $d_1 = 25 \text{ cm}$

Inner radius, $r_1 = 12.5 \text{ cm}$

$$r_1 = 0.125 \text{ m}$$

Radius, $r_2 = r_1 + \text{thickness of wall}$

$$= 0.125 + 0.02$$

$$r_2 = 0.145$$

Temperature of hot liquid, $T_a = 100^\circ\text{C} + 273$

$$T_a = 373 \text{ K}$$

Temperature of surrounding, $T_b = 20^\circ\text{C} + 273$

$$T_b = 293 \text{ K}$$

Inside heat transfer co-efficient, $h_a = 730 \text{ W/m}^2\text{K}$

Outside heat transfer co-efficient, $h_b = 12 \text{ W/m}^2\text{K}$

$$k_{\text{steel}} = 55 \text{ W/mK}$$

$$k_{\text{insulation}} = 0.22 \text{ W/mK}$$

To find:

Heat loss per metre length

Solution:

Heat flow through composite cylinder is given by

$$Q = \frac{\Delta T_{\text{overall}}}{R} \quad [\text{From Equn no. 1.48 or HMT Data book page No. 43 \& 44}]$$

Where

$$\Delta T = T_a - T_b$$

$$R = \frac{1}{2\pi L} \left[\frac{1}{h_a r_1} + \frac{\ln \left[\frac{r_2}{r_1} \right]}{k_1} + \frac{\ln \left[\frac{r_3}{r_2} \right]}{k_2} + \frac{1}{h_b r_3} \right]$$

$$\Rightarrow Q = \frac{T_a - T_b}{\frac{1}{2\pi L} \left[\frac{1}{h_a r_1} + \frac{\ln \left[\frac{r_2}{r_1} \right]}{k_1} + \frac{\ln \left[\frac{r_3}{r_2} \right]}{k_2} + \frac{1}{h_b r_3} \right]}$$

$$\Rightarrow \frac{Q}{L} = \frac{373 - 293}{\frac{1}{2\pi} \left[\frac{1}{730 \times .125} + \frac{\ln \left[\frac{.145}{.125} \right]}{55} + \frac{\ln \left[\frac{.195}{.145} \right]}{0.22} + \frac{1}{12 \times .195} \right]}$$

$$Q/L = 281.178 \text{ W/m}$$

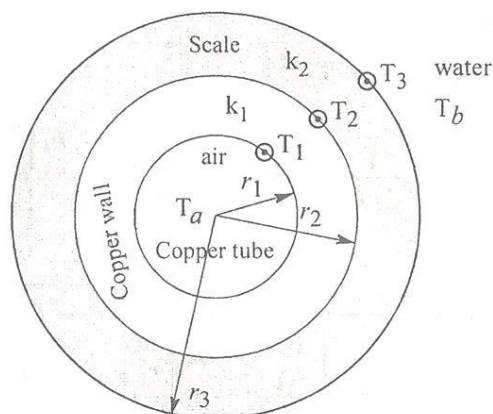
Result:

Heat transfer per metre length, $Q/L = 281.178 \text{ W/m}$.

5. Air at 90°C flows in a copper tube of 5 cm inner diameter with thermal conductivity 380 W/mK and with 0.7 cm thick wall which is heated from the outside by water at 120°C . a scale of 0.4 cm thick is deposited on the outer surface of the tube whose thermal conductivity is 1.82 W/mK. The air and water side unit surface conductance are $220 \text{ W/m}^2 \text{ K}$ and $3650 \text{ W/m}^2 \text{ K}$ respectively. Calculate

- Overall water to air transmittance
- Water to air heat exchange
- Temperature drop across the scale deposit.

Given:



Inner air temperature, $T_a = 90^\circ + 273$

$$T_a = 363 \text{ K}$$

Inner diameter of the copper, $d_1 = 5 \text{ cm}$

Radius, $r_1 = 2.5 \text{ cm}$

$$r_1 = 0.025 \text{ m}$$

Thermal conductivity, $k_1 = 380 \text{ W/mK}$

Outer radius of the copper, $r_2 = \text{inner radius} + \text{thickness of wall}$

$$r_2 = 0.025 + 0.007 \text{ m}$$

$$r_2 = 0.032 \text{ m}$$

$r_3 = r_2 + \text{thickness of scale}$

$$= 0.032 + 0.004$$

$$r_3 = 0.036 \text{ m}$$

Outside temperature of water, $T_b = 120^\circ\text{C} + 273$

$$= 393 \text{ K}$$

Thermal conductivity, $k_2 = 1.82 \text{ W/mK}$

Surface conductance of air, $h_a = 220 \text{ W/m}^2\text{K}$

Surface conductance of water, $h_b = 3650 \text{ W/m}^2\text{K}$

To find:

1. Overall heat transfer co-efficient, U
2. Water to air heat transfer, Q
3. Temperature drop across the scale deposit, $(T_3 - T_2)$

Solution:

Heat flow through composite cylinder is given by

$$Q = \frac{\Delta T_{overall}}{R} \quad [\text{From Equn no. (1.48) or HMT Data book page No. 43 \& 45}]$$

Where

$$\Delta T = T_a - T_b$$

$$R = \frac{1}{2\pi L} \left[\frac{1}{h_a r_1} + \frac{\ln \left[\frac{r_2}{r_1} \right]}{k_1} + \frac{\ln \left[\frac{r_3}{r_2} \right]}{k_2} + \frac{1}{h_b r_3} \right]$$

$$\Rightarrow Q = \frac{T_a - T_b}{\frac{1}{2\pi L} \left[\frac{1}{h_a r_1} + \frac{\ln \left[\frac{r_2}{r_1} \right]}{k_1} + \frac{\ln \left[\frac{r_3}{r_2} \right]}{k_2} + \frac{1}{h_b r_3} \right]}$$

$$\Rightarrow \frac{Q}{L} = \frac{363 - 393}{\frac{1}{2\pi} \left[\frac{1}{220 \times .025} + \frac{\ln \left[\frac{.032}{.025} \right]}{380} + \frac{\ln \left[\frac{.036}{.032} \right]}{1.82} + \frac{1}{3650 \times .036} \right]}$$

$Q/L = -739.79 \text{ W/m}$

[Negative sign indicates that heat flows from outside to inner side]

We know that,

$$\text{Heat transfer, } Q = UA \Delta T$$

Where

U – Overall heat transfer co-efficient

$$A - \text{Area} = 2 \pi r_3 L$$

$$\Delta T = T_a - T_b$$

$$\Rightarrow Q = U \times 2\pi_3 L \times (T_a - T_b)$$

$$\frac{Q}{L} = U \times 2\pi_3 \times (T_a - T_b)$$

$$\Rightarrow -739.79 = U \times 2 \times \pi \times 0.036 \times (363 - 393)$$

$$\Rightarrow \boxed{U = 109.01 \text{ W/m}^2\text{K}}$$

Overall heat transfer co-efficient, $U = 109.01 \text{ W/m}^2\text{K}$.

Interface temperatures

$$Q = \frac{\Delta T}{R} = \frac{T_a - T_b}{R} = \frac{T_a - T_1}{R_a} = \frac{T_1 - T_2}{R_1} = \frac{T_2 - T_3}{R_2} = \frac{T_3 - T_b}{R_b} \dots\dots\dots (1)$$

$$(1) \quad \Rightarrow Q = \frac{T_2 - T_3}{R_2}$$

Where

$$R_2 = \frac{1}{2\pi L} \left[\frac{\ln \frac{r_3}{r_2}}{k_2} \right]$$

$$\Rightarrow Q = \frac{T_2 - T_3}{\frac{1}{2\pi L} \left[\frac{\ln \frac{r_3}{r_2}}{k_2} \right]}$$

$$\Rightarrow \frac{Q}{L} = \frac{T_2 - T_3}{\frac{1}{2\pi} \left[\frac{\ln \frac{r_3}{r_2}}{k_2} \right]}$$

$$\Rightarrow -739.79 = \frac{T_2 - T_3}{\frac{1}{2\pi} \left[\frac{\ln \left[\frac{0.036}{0.032} \right]}{1.82} \right]}$$

$$\Rightarrow T_2 - T_3 = -7.6 \text{ K}$$

$$\Rightarrow T_3 - T_2 = 7.6 \text{ K}$$

$$\text{Temperature across the scale deposit, } T_3 - T_2 = 7.6 \text{ K}$$

Result:

1. Overall heat transfer co-efficient, $U = 109.01 \text{ W/m}^2\text{K}$
2. Heat exchange, $Q/L = -739.79 \text{ W/m}$
[Negative sign indicates that heat flows from outside to inner side]
3. Temperature drop across the scale deposit,
 $T_3 - T_2 = 7.6 \text{ K}$

6. An electrical wire of 10 m length and 1 mm diameter dissipates 200W in air at 25°C. The convection heat transfer co-efficient between the wire surface and air is 15W/m²K. The thermal conductivity of wire is 0.582 W/mK. Calculate the critical radius of insulation and also determine the temperature of the wire if it is insulated to the critical thickness of insulation.

Given:

Length of the wire, $L = 10 \text{ mm}$

Diameter of the wire, $d = 1 \text{ mm}$

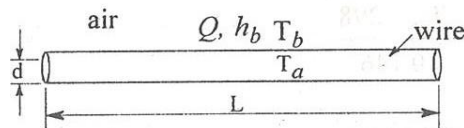
Radius of the wire, $r = 0.5 \text{ mm} = 0.5 \times 10^{-3} \text{ m}$

Heat transfer, $Q = 200 \text{ W}$

Surrounding temperature, $T_b = 25^\circ\text{C} + 273 = 298 \text{ K}$

Convection heat transfer co-efficient between the wire surface and air, $h_b = 15 \text{ W/m}^2\text{K}$.

Thermal conductivity of wire, $k = 0.582 \text{ W/mK}$



To find:

1. Critical radius of insulation, r_c
2. Temperature of wire, T_a

Solution:

We know that,

$$\text{Critical radius of insulation, } r_c = \frac{k}{n} \\ = \frac{0.582}{15}$$

$$r_c = 0.0388 \text{ m}$$

Heat transfer through an insulated wire when critical radius is used is given by

$$\Rightarrow Q = \frac{(T_a - T_b)}{\frac{1}{2\pi L} \left[\ln \left(\frac{r_c}{r_1} \right) + \frac{1}{h_b r_c} \right]}$$
$$200 = \frac{T_a - 298}{\frac{1}{2\pi \times 10} \left[\ln \left(\frac{0.0388}{0.0005} \right) + \frac{1}{15(0.0388)} \right]}$$
$$200 = \frac{T_a - 298}{0.146}$$
$$\Rightarrow T_a = 327.28 \text{ K}$$

Result:

1. Critical radius of insulation, $r_c = 0.0388\text{mm}$
2. Temperature of the wire, $T_a = 327.28 \text{ K}$ (or) 54.2°C .

7. A wire of 6 mm diameter with 2 mm thick insulation ($K = 0.11 \text{ W/mK}$). If the convective heat transfer co-efficient between the insulating surface heat transfer co-efficient between the insulating surface and air is $25 \text{ W/m}^2\text{K}$, find the critical thickness of insulation and also find the percentage of change in the heat transfer rate if the critical radius is used.

Given:

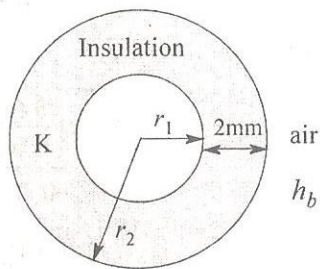
$$d_1 = 6 \text{ mm}$$

$$r_1 = 3 \text{ mm}$$

$$r_2 = r_1 + 2 = 3 + 2 = 5 \text{ mm} = 0.005 \text{ m}$$

$$k = 0.11 \text{ W / mK}$$

$$h_b = 25 \text{ W/m}^2\text{K}$$



Solution:

$$1. \text{ Critical radius, } r_c = \frac{k}{h} \quad [\text{From. No. (1.50)}]$$

$$r_c = \frac{0.11}{25} = 4.4 \times 10^{-3} \text{ m}$$

$$r_c = 4.4 \times 10^{-3} \text{ m}$$

$$\begin{aligned} \text{Critical thickness, } t_c &= r_c - r_1 \\ &= 4.4 \times 10^{-3} \text{ m} \end{aligned}$$

$$\text{Critical thickness, } t_c = 1.4 \times 10^{-3} \text{ m (or) } 1.4 \text{ mm}$$

2. Heat transfer through an insulated wire is given by

$$\Rightarrow Q_1 = \frac{(T_a - T_b)}{\frac{1}{2\pi L} \left[\frac{\ln \left[\frac{r_2}{r_1} \right]}{k_1} + \frac{1}{h_b r_2} \right]}$$

$$= \frac{2\pi(T_a - T_b)}{\frac{\ln \left[\frac{0.005}{0.003} \right]}{0.11} + \frac{1}{25 \times 0.005}}$$

[From HMT data book page no. 43 & 45]

$$Q_1 = \frac{2\pi L(T_a - T_b)}{12.64}$$

Heat flow through an insulated wire when critical radius is used is given by

$$\Rightarrow Q_2 = \frac{(T_a - T_b)}{\frac{1}{2\pi L} \left[\frac{\ln \left[\frac{r_c}{r_1} \right]}{k_1} + \frac{1}{h_b r_c} \right]}$$

$$= \frac{2\pi L(T_a - T_b)}{\frac{\ln \left[\frac{4.4 \times 10^{-3}}{0.003} \right]}{0.11} + \frac{1}{25 \times 4.4 \times 10^{-3}}}$$

$$Q_2 = \frac{2\pi L(T_a - T_b)}{12.572}$$

∴ Percentage of increase in heat flow by using

$$\text{Critical radius} = \frac{Q_2 - Q_1}{Q_1} \times 100$$

$$= \frac{\frac{1}{12.57} - \frac{1}{12.64}}{\frac{1}{12.64}} \times 100$$

$$= 0.55\%$$

Result:

1. Critical thickness, $t_c = 1.4 \times 10^{-3}$
2. Percentage of increase in heat transfer by using critical radius = 0.55%

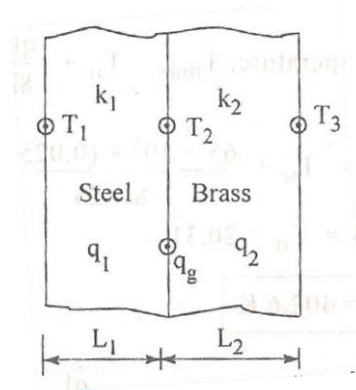
8. An electric current is passed through a composite wall made up of two layers. First layer is steel of 10 cm thickness and second layer is brass of 8 cm thickness. The outer surface temperature of steel and brass are maintained at 120°C and 65°C respectively. Assuming that the contact between two slab is perfect and the heat generation is 1, 65,000 W/m³.

Determine:

1. Heat flux through the outer surface of brass slab
2. Interface temperature.

Take k for steel is 45 W/mK. K for brass is 80 W/mK.

Given:



Thickness of steel, $L_1 = 10 \text{ cm} = 0.10 \text{ m}$

Thickness of brass, $L_2 = 8 \text{ cm} = 0.08 \text{ m}$

Surface temperature of steel, $T_1 = 120^\circ\text{C} + 273 = 393 \text{ K}$

Outside surface temperature of brass, $T_3 = 25^\circ\text{C} + 273 = 293 \text{ K}$

Heat generation, $q_g = 1, 65,000 \text{ W/m}^3$

$$k_1 = 45 \text{ W/mK}$$

$$k_2 = 80 \text{ W/mK}$$

To find:

1. Heat flux through the surface of the brass slab, q_2
2. Interface temperature, T_2 .

Solution:

Let

q_1 – Heat flux through the surface of the steel slab

q_2 – Heat flux through the surface of the brass slab

Heat generation $q_g = q_1 + q_2$ (1)

Heat transfer through steel,

$$Q_1 = \frac{\Delta T}{R}$$
$$Q_1 = \frac{T_1 - T_2}{\frac{L_1}{k_1 A}} \quad \left[\because R = \frac{L}{kA} \right]$$

Let interface temperature T_2 is greater than T_1 . So,

$$Q_1 = \frac{T_2 - T_1}{\frac{L_1}{k_1 A}} \quad \text{..... (2)}$$

Heat transfer through brass is given by

$$Q_2 = \frac{\Delta T}{R}$$
$$Q_2 = \frac{T_2 - T_3}{\frac{L_2}{k_2 A}} \quad \text{..... (3)}$$

Total heat transfer [Adding (2) + (3)]

$$Q_1 = Q_1 + Q_2$$

$$Q_1 = \frac{T_2 - T_1}{\frac{L_1}{k_1 A}} + \frac{T_2 - T_3}{\frac{L_2}{k_2 A}}$$

$$Q / A = \frac{T_2 - T_1}{\frac{L_1}{k_1}} + \frac{T_2 - T_3}{\frac{L_2}{k_2}}$$

Heat flux (or) Heat generation

$$qg = Q / A = \frac{T_2 - T_1}{\frac{L_1}{k_1}} + \frac{T_2 - T_3}{\frac{L_2}{k_2}}$$

$$q_g = \frac{T_2 - 393}{\frac{0.10}{45}} + \frac{T_2 - 338}{\frac{0.08}{80}}$$

$$\Rightarrow 1,65,000 = \frac{T_2 - 393}{2.2 \times 10^{-3}} + \frac{T_2 - 338}{1 \times 10^{-3}}$$

$$\frac{T_2}{2.2 \times 10^{-3}} - \frac{393}{2.2 \times 10^{-3}} + \frac{T_2}{1 \times 10^{-3}} - \frac{338}{1 \times 10^{-3}}$$

$$= T_2 [454.54 + 1000] - 1,78,636.3 - 338000$$

$$= T_2 [1454.54] - [5,16,636]$$

$$\Rightarrow 1,65,000 + 5,16,636 = T_2 [1454.5]$$

$$\Rightarrow T_2 = 468.6K$$

Interface temperature, $T_2 = 468.6K$

Heat transfer through steel, is given by

$$Q_1 = \frac{T_2 - T_1}{\frac{L_1}{k_1 A}}$$

$$Q_1 / A = \frac{T_2 - T_1}{\frac{L_1}{k_1}}$$

$$\Rightarrow q_1 = \frac{468.6 - 393}{\frac{0.10}{45}}$$

$q_1 = 34020 \text{ W/m}^2$

From equation (1)

$$\text{Heat generation, } q_g = q_1 + q_2$$

$$\Rightarrow 1,65,000 = 34,020 + q_2$$

$$\Rightarrow q_2 = 1,65,000 - 34,020$$

$$q_2 = 1,30,980 \text{ W/m}^2$$

Heat flux through the

Surface of the brass slab, $q_2 = 1,30,980 \text{ W/m}^2$

Result:

(i) $q_2 = 1,30,980 \text{ W/m}^2$

(ii) $T_2 = 468.6 \text{ K}$.

9. A copper wire of 40 mm diameter carries 250 A and has a resistance of $0.25 \times 10^{-4} \Omega$ cm/length surface temperature of copper wire is 250°C and the ambient air temperature is 10°C . If the thermal conductivity of the copper wire is 175 W/mK , calculate

1. Heat transfer co-efficient between wire surface and ambient air.
2. Maximum temperature in the wire.

Given:

Diameter, $d = 40 \text{ mm} = 0.040 \text{ m}$

Radius, $r = 20 \text{ mm} = 0.020 \text{ m}$

Current, $I = 250 \text{ A}$.

Resistance, $R = 0.25 \times 10^{-4} \Omega \text{ cm/length}$

Surface temperature, $T_w = 205^\circ\text{C} + 273 = 523 \text{ K}$

Ambient air temperature, $T_\infty = 10^\circ\text{C} + 273 = 283 \text{ K}$

Thermal conductivity, $k = 175 \text{ W/mK}$

To find:

1. Heat transfer co-efficient, h
2. Maximum temperature, $T_{\max}$.

Solution:

$$\begin{aligned}\text{Heat transfer, } Q &= I^2 R \\ &= (250)^2 \times (0.25 \times 10^{-4}) \\ &= 1.562 \text{ W/cm} \\ &= 1.56 \times 10^2 \text{ W/m} \\ &= 156 \text{ W/m}\end{aligned}$$

We know that,

$$\begin{aligned}\text{Heat generated, } q &= \frac{Q}{V} = \frac{156}{\pi \times r^2 \times L} \\ q &= \frac{156}{\pi \times (0.020)^2 \times 1}\end{aligned}$$

$$q = 124140 \text{ W/m}^3$$

We know that,

Maximum temperature

$$T_{\max} = T_w + \frac{qt^2}{4k} \quad [\text{From Equn no. (1.62)}]$$

$$\begin{aligned}&= 523 + \frac{124140 \times (0.020)^2}{4 \times 175} \\ &= 523.07 \text{ K.}\end{aligned}$$

$$T_{\max} = 523.07 \text{ K.}$$

We know that,

$$\text{Surface temperature, } T_w = T_\infty + \frac{rq}{2h} \quad [\text{From Equn no. (1.65)}]$$

$$523 = 283 + \frac{0.020 \times 124140}{2 \times h}$$

==> $h = 5017 \text{ W/m}^2\text{K}$

Result:

1. Heat transfer co-efficient, $h = 5.17 \text{ W/m}^2\text{K}$
2. Maximum temperature, $T_{\max} = 523.07 \text{ K}$.

10. An aluminum alloy fin of 7 mm thick and 50 mm long protrudes from a wall, which is maintained at 120°C . The heat transfer coefficient and conductivity of the fin material are $140 \text{ W/m}^2\text{K}$ and 55 W/mK respectively. Determine

- 1. Temperature at the end of the fin.**
- 2. Temperature at the middle of the fin.**
- 3. Total heat dissipated by the fin.**

Given:

Thickness, $t = 7 \text{ mm} = 0.007 \text{ m}$

Length, $L = 50 \text{ mm} = 0.050 \text{ m}$

Base temperature, $T_b = 120^\circ\text{C} + 273 = 393 \text{ K}$

Ambient temperature, $T_\infty = 22^\circ + 273 = 295 \text{ K}$

Heat transfer co-efficient, $h = 140 \text{ W/m}^2\text{K}$

Thermal conductivity, $k = 55 \text{ W/mK}$.

To find:

1. Temperature at the end of the fin.
2. Temperature at the middle of the fin
3. Total heat dissipated by the fin.

Solution:

Since the length of the fin is 50 mm, it is treated as short fin. Assume end is insulated.

We know that,

Temperature distribution [Short fin, end insulated]

$$\frac{T - T_{\infty}}{T_b - T_{\infty}} = \frac{\cosh m[L - x]}{\cosh(mL)} \quad \dots\dots (1)$$

[From HMT data book page no. 49]

i) Temperature at the end of the fin, Put $x = L$

$$\begin{aligned} (1) \implies \frac{T - T_{\infty}}{T_b - T_{\infty}} &= \frac{\cosh m[L - L]}{\cosh(mL)} \\ \implies \frac{T - T_{\infty}}{T_b - T_{\infty}} &= \frac{1}{\cosh(mL)} \end{aligned} \quad \dots\dots (2)$$

Where

$$m = \sqrt{\frac{hP}{kA}}$$

P = Perimeter = $2 \times L$ (Approx)

$$= 2 \times 0.050$$

$P = 0.1 \text{ m}$

A – Area = Length $\times$ thickness = 0.050×0.007

$A = 3.5 \times 10^{-4} \text{ m}^2$

$$m = \sqrt{\frac{hP}{kA}}$$

$$= \sqrt{\frac{140 \times 0.1}{55 \times 3.5 \times 10^{-4}}}$$

$m = 26.96 \text{ m}^{-1}$

$$\begin{aligned}
 (2) \implies \frac{T - T_{\infty}}{T_b - T_{\infty}} &= \frac{1}{\cosh(26.9 \times 0.050)} \\
 \implies \frac{T - T_{\infty}}{T_b - T_{\infty}} &= \frac{1}{2.05} \\
 \implies \frac{T - 295}{393 - 295} &= \frac{1}{2.05} \\
 \implies T - 295 &= 47.8 \\
 \implies \boxed{T = 342.8 \text{ K}}
 \end{aligned}$$

Temperature at the end of the fin, $T_{x=L} = 342.8 \text{ K}$

ii) Temperature at the middle of the fin.

Put $x = L/2$ in Equation (1)

$$\begin{aligned}
 (1) \implies \frac{T - T_{\infty}}{T_b - T_{\infty}} &= \frac{\cosh m[L - L/2]}{\cosh(mL)} \\
 \implies \frac{T - T_{\infty}}{T_b - T_{\infty}} &= \frac{\cosh 26.9 \left[0.050 - \frac{0.050}{2} \right]}{\cosh[26.9 \times (0.050)]} \\
 \implies \frac{T - 295}{393 - 295} &= \frac{1.234}{2.049} \\
 \implies \frac{T - 295}{393 - 295} &= 0.6025
 \end{aligned}$$

$$\boxed{T = 354.04 \text{ K}}$$

Temperature at the middle of the fin

$$\boxed{T_{x=L/2} = 354.04 \text{ K}}$$

iii) Total heat dissipated

[From HMT data book page no. 49]

$$\begin{aligned}
 Q &= (hPkA)^{1/2} (T_b - T_{\infty}) \tanh(mL) \\
 &= [140 \times 0.1 \times 55 \times 3.5 \times 10^{-4}]^{1/2} \times (393 - 295) \times \tanh(26.9 \times 0.050)
 \end{aligned}$$

$$\boxed{Q = 44.4 \text{ W}}$$

Result:

1. Temperature at the end of the fin, $T_{x=L} = 342.8K$
2. Temperature at the middle of the fin, $T_{x=L/2} = 354.04K$
3. Total heat dissipated, $Q = 44.4 \text{ W}$

11. Ten thin brass fins ($k=100 \text{ W/mK}$), 0.75 mm thick are placed axially on a 1 m long and 60 mm diameter engine cylinder which is surrounded by 27°C . The fins are extended 1.5 m from the cylinder surface and the heat transfer co-efficient between cylinder and atmospheric air is $15 \text{ W/m}^2\text{K}$. Calculate the rate of heat transfer and the temperature at the end of fins when the cylinder surface is at 160°C .

Given:

Number of fins = 10

Thermal conductivity, $k = 100 \text{ W/mK}$

Thickness of the fin, $t = 0.75 \times 10^{-3} \text{ m}$

Length of engine cylinder, $L_{cy} = 1 \text{ m}$

Diameter of the cylinder, $d = 60 \text{ mm} = 0.060 \text{ m}$

Atmosphere temperature, $T_\infty = 27^\circ\text{C} + 273 = 300 \text{ K}$

Length of the fin, $L_f = 1.5 \text{ cm} = 1.5 \times 10^{-2} \text{ m}$

Heat transfer co-efficient, $h = 15 \text{ W/m}^2\text{K}$

Cylinder surface temperature

Or

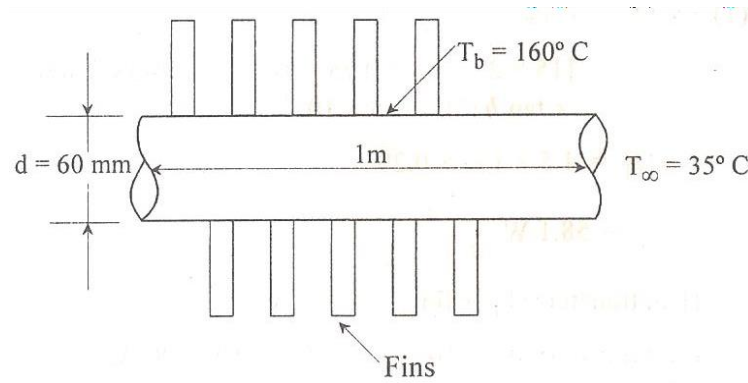
base temperature, $T_b = 160^\circ\text{C} + 273 = 433 \text{ K}$

To find:

1. Rate of heat transfer Q
2. Temperature at the end of the fin

Solution:

Length of the fin is 1.5 cm . So, this is short fin. Assuming that the fin end is insulated



We know that,

$$\text{Heat transferred, } Q = (hPkA)^{1/2} (T_b - T_\infty) \tanh(mL) \quad \dots\dots\dots (1)$$

[From HMT data book page no.49]

Where

P – Perimeter = 2 × Length of the cylinder

$$= 2 \times 1$$

$$\boxed{P = 2 \text{ m}}$$

A – Area = length of the cylinder × thickness

$$= 1 \times 0.75 \times 10^{-3} \text{ m}$$

$$\boxed{A = 1 \times 0.75 \times 10^{-3}}$$

$$m = \sqrt{\frac{hP}{kA}}$$

$$= \sqrt{\frac{15 \times 2}{100 \times 0.75 \times 10^{-3}}}$$

$$\boxed{20 = \text{m}^{-1}}$$

$$\begin{aligned}
 (1) \Rightarrow Q_1 &= (hPkA)^{1/2} (T_b - T_\infty) \tanh(mL_f) \\
 &= [15 \times 2 \times 100 \times 0.75 \times 10^{-3}]^{1/2} \times (433 - 300) \times \tanh(20 \times 1.5 \times 10^{-2}) \\
 Q_1 &= 1.5 \times 133 \times 0.29 \\
 Q_1 &= 58.1 \text{ W}
 \end{aligned}$$

Heat transferred per fin = 58.1 W

Heat transferred for 10 fins = $58.1 \times 10 = 581 \text{ W}$

$$\boxed{Q_1 = 581 \text{ W}} \quad \dots\dots\dots (2)$$

Heat transfer from un finned surface due to convection is

$$\begin{aligned}
 Q_2 &= hA\Delta T \\
 &= h \times (\pi d L_{cy} - 10 \times t \times L_f) (T_b - T_\infty)
 \end{aligned}$$

[$\because$ Area of unfinned surface = Area of cylinder – Area of fin]

$$= 15 \times [(\pi \times 0.060 \times 1) - (10 \times 0.75 \times 10^{-3} \times 1.5 \times 10^{-2})] [433 - 300]$$

$$\boxed{Q_2 = 375.8 \text{ W}} \quad \dots\dots\dots (3)$$

So, total heat transfer, $Q = Q_1 + Q_2$

$$Q = 581 + 375.8$$

$$\boxed{\text{Total heat transfer, } Q = 956.8 \text{ W}}$$

We know that,

Temperature distribution [short fin, end insulated]

$$\Rightarrow \frac{T - T_\infty}{T_b - T_\infty} = \frac{\cosh m[L_f - x]}{\cosh mL_f}$$

[From HMT data book page no.49]

We need temperature at the end of fin, so, put $x = L$

$$\begin{aligned}
\Rightarrow \frac{T - T_{\infty}}{T_b - T_{\infty}} &= \frac{\cosh m[L - L]}{\cosh(mL)} \\
\Rightarrow \frac{T - T_{\infty}}{T_b - T_{\infty}} &= \frac{1}{\cosh(20 \times 1.5 \times 10^{-2})} \\
&= \frac{1}{0.95} \\
\Rightarrow T - T_{\infty} &= \frac{T_b - T_{\infty}}{0.95} \\
\Rightarrow T = T_{\infty} + \frac{T_b - T_{\infty}}{0.95} \\
&= 300 + \frac{433 - 300}{0.95}
\end{aligned}$$

$T = 440 \text{ K}$

Result:

1. Heat transfer, $Q = 956.8 \text{ W}$
2. Temperature at the end of the fin, $T = 440 \text{ K}$.

12. An aluminum plate ($k = 160 \text{ W/m}^\circ\text{C}$, $\rho = 2790 \text{ kg/m}^3$, $C_p = 0.88 \text{ KJ/kg}^\circ\text{C}$) of thickness $L = 3 \text{ cm}$ and at a uniform temperature of 225°C is suddenly immersed at time $t = 0$ in a well stirred fluid maintained at a constant temperature $T_{\infty} = 25^\circ\text{C}$. Take $h = 320 \text{ W/m}^2\text{C}$. Determine the time required for the centre of the plate to reach 50°C .

Given:

Thermal conductivity of aluminum, $k = 160 \text{ W/m}^\circ\text{C}$

Density, $\rho = 2790 \text{ kg/m}^3$

Specific heat, $C_p = 0.88 \text{ KJ/kg}^\circ\text{C} = 0.88 \times 10^3 \text{ J/kg}^\circ\text{C}$

Thickness, $L = 3 \text{ cm} = 0.03 \text{ m}$

Initial temperature, $T_0 = 225^\circ\text{C} + 273 = 498 \text{ K}$

Final temperature, $T_{\infty} = 25^\circ\text{C} + 273 = 323 \text{ K}$

Heat transfer co-efficient, $h = 320 \text{ W/m}^2\text{C}$

To find:

Time (t) required to reach 50°C .

Solution:

We know that, For slab,

$$\begin{aligned} \text{Characteristic length, } L_c &= \frac{L}{2} \\ &= \frac{0.03}{2} \end{aligned}$$

$$L_c = 0.015 \text{ m}$$

$$\begin{aligned} \text{Biot number } B_i &= \frac{hL_e}{k} \\ &= \frac{320 \times 0.015}{160} \end{aligned}$$

$$B_i = 0.03 < 0.1$$

Biot number value is less than 0.1. So, this is lumped heat analysis type problem.

For lumped parameter system.

$$\frac{T - T_\infty}{T_0 - T_\infty} = e^{\left[\frac{-hA}{C_p \times V \times \rho} t \right]} \quad [\text{From HMT data book page no.57}]$$

We know that,

$$\text{Characteristics length, } L_c = \frac{V}{A}$$

$$\begin{aligned} (1) \implies \frac{T - T_\infty}{T_0 - T_\infty} &= e^{\left[\frac{-hA}{C_p \times V \times \rho} t \right]} \\ \implies \frac{323 - 298}{498 - 298} &= e^{\left[\frac{-320 \times t}{0.88 \times 10^3 \times 0.015 \times 2790} \right]} \\ \implies \ln(0.125) &= \frac{-320 \times t}{0.88 \times 10^3 \times 0.015 \times 2790} \\ \implies -2.079 &= -0.00868t \end{aligned}$$

$\implies$

$$t = 239.26 \text{ s}$$

Result:

Time required to reach 50°C is 239.26 s.

13. A steel ball (specific heat = 0.46 kJ/kgK. and thermal conductivity = 35 W/mK) having 5 cm diameter and initially at a uniform temperature of 450°C is suddenly placed in a control environment in which the temperature is maintained at 100°C. Calculate the time required of the ball to attained a temperature of 150°C.

Take $h = 10 \text{ W/m}^2\text{K}$

Given:

Specific heat, $C_p = 0.46 \text{ kJ/kg K} = 460 \text{ J/kg K}$

Thermal conductivity, $k = 35 \text{ W/mK}$

Diameter of the sphere, $D = 5 \text{ cm} = 0.05 \text{ m}$

Radius of the sphere, $R = 0.025 \text{ m}$

Initial temperature, $T_0 = 450^\circ\text{C} + 273 = 723 \text{ K}$

Final temperature, $T_\infty = 100^\circ\text{C} + 273 = 373 \text{ K}$

Intermediate temperature, $T = 150^\circ\text{C} + 273 = 423 \text{ K}$

Heat transfer co-efficient, $h = 10 \text{ W/m}^2\text{K}$

To find:

Time required for the ball to reach 150°C

Solution:

Density of steel is 7833 kg/m^3

$$\rho = 7833 \text{ kg / m}^3$$

For sphere,

$$\begin{aligned}\text{Characteristic length, } L_c &= \frac{R}{3} \\ &= \frac{0.025}{3}\end{aligned}$$

$$L_c = 8.33 \times 10^{-3} \text{ m}$$

We know that,

$$\begin{aligned} \text{Biot number, } B_i &= \frac{hL_e}{k} \\ &= \frac{10 \times 8.3 \times 10^{-3}}{35} \\ B_i &= 2.38 \times 10^{-3} < 0.1 \end{aligned}$$

Biot number value is less than 0.1. So, this is lumped heat analysis type problem.

For lumped parameter system,

$$\frac{T - T_\infty}{T_0 - T_\infty} = e^{\left[\frac{-hA}{C_p \times V \times \rho} t \right]} \quad \dots\dots (1)$$

[From HMT data book page no.57]

We know that,

$$\text{Characteristics length, } L_c = \frac{V}{A}$$

$$\begin{aligned} (1) \implies \frac{T - T_\infty}{T_0 - T_\infty} &= e^{\left[\frac{-hA}{C_p \times V \times \rho} t \right]} \\ \implies \frac{423 - 373}{723 - 373} &= e^{\left[\frac{-10}{460 \times 8.33 \times 10^{-3} \times 7833} t \right]} \\ \implies \ln \frac{423 - 373}{723 - 373} &= \frac{-10}{460 \times 8.33 \times 10^3 \times 7833} \times t \\ \implies &\boxed{t = 5840.54 \text{ s}} \end{aligned}$$

Result:

Time required for the ball to reach 150°C is 5840.54s.

14. A large wall 2 cm thick has uniform temperature 30°C initially and the wall temperature is suddenly raised and maintained at 400°C. Find

1. The temperature at a depth of 0.8 cm from the surface of the wall after 10 s.

2. *Instantaneous heat flow rate through that surface per m² per hour.*

Take $\alpha = 0.008 \text{ m}^2/\text{hr}$, $k = \text{W}/\text{m}^\circ\text{C}$.

Given:

Thickness, $L = 2 \text{ cm} = 0.02 \text{ m}$

Initial temperature, $T_i = 30^\circ\text{C} + 273 = 303 \text{ K}$

Surface temperature, $T_0 = 400^\circ\text{C} + 273 = 673 \text{ K}$

Thermal diffusivity, $\alpha = 0.008 \text{ m}^2/\text{h}$
 $= 2.22 \times 10^{-6} \text{ m}^2/\text{s}$

Thermal conductivity, $k = 6 \text{ W}/\text{m}^\circ\text{C}$.

Case (i)

Depth, $x = 0.8 \text{ cm} = 0.8 \times 10^{-2} \text{ m}$
 $= 0.008 \text{ m}$

Time, $t = 10 \text{ s}$

Case (ii)

Time, $t = 1 \text{ h} = 3600 \text{ s}$

To find:

1. Temperature (T_x) at a depth of 0.8 cm from the surface of the wall after 10s.
2. Instantaneous heat flow rate (q_x) through that surface per hour.

Solution:

In this problem heat transfer co-efficient h is not given. So take it as ∞ . i.e., $h \rightarrow \infty$.

We know that,

$$\text{Biot number, } B_i = \frac{hL_e}{k}$$

$$h = \infty$$

$h = \infty$

$\Rightarrow$

Bi value is ∞ . So, this is semi infinite solid type problem.

Case (i)

For semi infinite solid,

$$\frac{T_x - T_0}{T_i - T_0} = \text{erf} \left[\frac{x}{2\sqrt{\alpha t}} \right]$$

[From HMT data book page no.58]

$$\frac{T_x - T_0}{T_i - T_0} = \text{erf}(Z) \quad \dots\dots (1)$$

Where,

$$Z = \frac{x}{2\sqrt{\alpha t}}$$

Put $x = 0.008 \text{ m}$, $t = 10 \text{ s}$, $\alpha = 2.22 \times 10^{-6} \text{ m}^2/\text{s}$.

$$\Rightarrow Z = \frac{0.008}{2\sqrt{2.22 \times 10^{-6} \times 10}}$$

$$\boxed{Z = 0.848}$$

$Z = 0.848$, corresponding erf (Z) is 0.7706

$$\Rightarrow \boxed{\text{erf}(Z) = 0.7706}$$

[Refer HMT data book page no. 59]

$$\begin{aligned}
 (1) \implies \frac{T_x - T_0}{T_i - T_0} &= 0.7706 \\
 \implies \frac{T_x - 673}{303 - 673} &= 0.7706 \\
 \implies \frac{T_x - 673}{-370} &= 0.7706
 \end{aligned}$$

$$T_x = 387.85K$$

Case (ii)

Instantaneous heat flow

$$q_x = \frac{k(T_0 - T_i)}{\sqrt{\alpha \pi t}} e^{\left[\frac{-x^2}{4\alpha t} \right]} \quad [\text{From HMT data book page no. 58}]$$

$t = 36000 \text{ s}$ (Given)

$$\implies q_x = \frac{6(673 - 303)}{\sqrt{\pi \times 2.22 \times 10^{-6} \times 3600}} e^{\left[\frac{-(0.008)^2}{4 \times 2.22 \times 10^{-6} \times 3600} \right]}$$

$$q_x = 13982.37 \text{ W / m}^2$$

Result:

Intermediate temperature, $T_x = 387.85 \text{ K}$

Heat flux, $q_x = 13982.37 \text{ W/m}^2$

15. A semi infinite slab of aluminum is exposed to a constant heat flux at the surface of 0.25 MW/m^2 . Initial temperature of the slab is 25°C . Calculate the surface temperature after 10 minutes and also find the temperature at a distance of 30 cm from the surface after 10 minutes.

Given:

Heat flux, $q_0 = 0.25 \text{ MW/m}^2$

$$q_0 = 0.25 \times 10^6 \text{ W/m}^2$$

Initial temperature, $T_i = 25^\circ\text{C} + 273 = 298\text{ K}$

Distance, $x = 30\text{ cm} = 0.30\text{ m}$

Time, $t = 10\text{ minutes} = 600\text{ s}$

To find:

1. Surface temperature (T_0) after 10 minutes.
2. Temperature (T_x) at a distance of 30 cm from the surface

Solution:

$$\text{Heat flux, } q_0 = \frac{k[T_0 - T_i]}{\sqrt{\alpha\pi t}} \quad \dots\dots (1)$$

[From HMT data book page no. 58]

[HMT data book page no. 1]

Properties of aluminum

Thermal diffusivity, $\alpha = 84.18 \times 10^{-6}\text{ m}^2/\text{s}$

Thermal conductivity, $k = 204.2\text{ W/mK}$.

$$(1) \Rightarrow 0.25 \times 10^6 = \frac{204.2(T_0 - 298)}{\sqrt{\pi \times 84.18 \times 10^{-6} \times 600}}$$

(ii) For semi infinite solid,

$T_0 = 785.68\text{ K}$

$$\frac{T_x - T_0}{T_i - T_0} = \text{erf} \left[\frac{x}{2\sqrt{\alpha t}} \right] \quad [\text{From HMT data book page no. 58}]$$

$$\frac{T_x - T_0}{T_i - T_0} = \text{erf}(Z) \quad \dots\dots (2)$$

Where

$$Z = \frac{x_0}{2\sqrt{\alpha t}}$$

$$\implies Z = \frac{0.30}{2\sqrt{4.18 \times 10^{-6} \times 600}}$$

$$Z = 0.667$$

$Z = 0.667$, corresponding erf (Z) is 0.65663

[From HMT data book page no. 59]

$$\text{erf}(Z) = 0.65663$$

$$(2) \implies \frac{T_x - T_0}{T_i - T_0} = 0.65663$$

$$\implies \frac{T_x - 785.68}{298 - 785.68} = 0.65663$$

$$\implies T_x = 465.45 \text{ K}$$

Temperature at a distance at 30 cm is 465.45 K

Result:

1. Surface temperature, $T_0 = 785.65 \text{ K}$
2. Temperature at a distance of 30 cm, $T_x = 465.45 \text{ K}$